

FEP-SSL: A Privacy-Preserving Semi-Supervised Learning Framework for Parkinson’s Disease Classification

Md. Shakil Ahmed¹, Shakila Shafiq², M Shamim Kaise^{1,*}, Md. Sazzadur Rahman¹, A. S. M. Sanwar Hosen^{3,*}, Youngbae Hwang⁴

1 Institute of Information Technology, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh, **2** Department of Computer Science & Engineering, National Institute of Textile Engineering and Research, Savar. Dhaka 1350, Bangladesh, **3** Department of Artificial Intelligence and Big Data, Woosong University, Daejeon 34606, South Korea, **4** Department of Intelligent Systems & Robotics, Chungbuk National University, Cheongju 28644, Republic of Korea.

* mskaiser@juniv.edu (MSK); sanwar@wsu.ac.kr (ASMSH).

Abstract

Background

Early detection of Parkinson’s Disease (PD) remains a significant challenge for healthcare service providers due to the limited labeled, class-imbalanced and strict privacy regulations that limit data sharing among healthcare institutions. These constraints hinder the development of robust Machine Learning (ML) models for diagnosis using distributed clinical datasets.

Objective

This paper introduces a privacy-preserving Semi-Supervised Learning (SSL) system that integrates Federated Learning (FL) with ensemble-guided pseudo-labeling to improve PD categorization using speech-derived features.

Methods

The proposed system, Federated Ensemble-Guided Pseudo-Labeling-based Semi-Supervised Learning framework (FEP-SSL), effectively combines SSL with FL optimization to utilize both labeled and unlabeled data distributed across multiple clients. The preprocessing stage includes label encoding, handling class imbalance using the Synthetic Minority Over-sampling Technique, and feature normalization. At the local level, three boosting model Extreme Gradient Boosting, Gradient Boosting, and Light Gradient Boosting Machine, generate pseudo-labels through a confidence-aware majority voting mechanism, thereby reducing noise propagation. At the global level, model parameters are aggregated using the Federated Averaging approach, enabling collaborative learning without sharing raw patient data.

Results and conclusion

The experimental evaluation on two benchmark PD speech datasets demonstrates strong performance. The federated global model achieves 97.14% accuracy on Dataset 1 and 95.50% on Dataset 2, outperforming centralized and baseline SSL approaches. Ablation studies reveal that MFCC, TQWT, and temporal speech features significantly enhance classification performance. Furthermore, feature importance analysis using SHAP highlights the contribution of key speech characteristics. The proposed framework reduces labeling costs, mitigates class imbalance, and preserves data privacy, demonstrating its effectiveness as a scalable and reliable solution for PD diagnosis using speech-based data. The findings indicate its viability as a scalable, dependable method

for diagnosing PD using speech-based data.

Keywords: Parkinson's Disease Classification, Semi-Supervised Learning, Pseudo-Labeling, Federated Learning, Privacy-Preserving.

1 Introduction

Parkinson’s Disease (PD) is a progressive neurodegenerative disorder characterized by the gradual loss of dopaminergic neurons in the substantia nigra of the midbrain, a region essential for motor control and coordination [1,2]. It is the second most prevalent neurological disorder worldwide, affecting more than 10 million individuals [3,4]. The depletion of dopamine disrupts normal motor functioning, leading to characteristic symptoms such as tremor, rigidity, bradykinesia, and postural instability. Beyond motor dysfunction, PD is frequently accompanied by a variety of non-motor manifestations, including cognitive deterioration, sleep-related issues, autonomic irregularities, and emotional disturbances. Detecting the disease at an early stage is crucial for aimed at delaying its progression, improving patients’ lives, and lowering long-term medical costs. However, accurate early-stage diagnosis remains challenging, as current clinical practices largely rely on neurological examinations and longitudinal assessments, with no definitive laboratory test available for early confirmation of PD.

Researchers have explored several non-invasive diagnostic approaches to enhance early detection of PD, including medical imaging, biosignal analysis, and speech-based assessment. Imaging techniques such as Magnetic Resonance Imaging (MRI) [5] and Positron Emission Tomography [6] can identify structural and functional abnormalities in the brain associated with PD. However, these techniques depend on specialized equipment, are expensive, and are often not readily accessible, particularly in resource-limited healthcare settings. Biosignal-based approaches, such as wearable sensors for monitoring gait or tremor patterns, provide valuable physiological insights. Nevertheless, they typically require patients to use dedicated devices, which may be inconvenient for continuous or long-term monitoring. In contrast, speech analysis has emerged as an effective, non-invasive, and cost-efficient alternative for PD detection. A significant proportion of patients exhibit speech impairments even in the early stages, including reduced vocal intensity, monotonic pitch, imprecise articulation, and increased jitter and shimmer. These acoustic variations can be captured using low-cost recording devices, enabling remote monitoring and large-scale data collection, thereby making speech data highly suitable for computational analysis and automated diagnostic systems.

Recent developments in Machine Learning (ML) have demonstrated promising performance in PD classification using speech and other biomedical signals. With the rapid growth of digital healthcare data and interconnected medical systems, Artificial Intelligence (AI) has become increasingly integral to clinical decision-support applications. Prior studies highlight the effectiveness of AI-driven models in neurological disease detection [7–9], disease management [10–13], elderly care systems [14,15], and biomedical data mining [16,17]. Despite these advancements, the performance of existing ML-based diagnostic models is often hindered by two key issues: the lack of labeled medical data and stringent data privacy requirements. Conventional supervised learning methods rely on large labeled dataset, which are not easy to obtain in clinical environments due to the need for expert knowledge, time-intensive labeling processes, and high associated costs. Moreover, medical data are typically distributed across multiple institutions and governed by strict privacy regulations, making centralized data aggregation impractical. These challenges give rise to serious concerns regarding to patient privacy, data protection, and the ownership of institutional information. Therefore, developing advanced learning frameworks that can effectively leverage limited labeled data while preserving privacy in decentralized healthcare settings remains a critical challenge for AI-based PD diagnosis.

To overcome the challenges of limited labeled data and restricted data sharing in healthcare environments, recent research has focused on alternative learning paradigms. Semi-Supervised Learning (SSL) [18] has emerged as an effective approach to mitigate label scarcity by leveraging both labeled and unlabeled data during training. For instance, pseudo-labeling techniques allow models to iteratively assign labels to unlabeled samples, thereby reducing dependence on large annotated datasets. Concurrently, Federated Learning (FL) [19] has emerged as a privacy-aware distributed learning strategy that allows multiple institutions to jointly train models while ensuring that sensitive patient data remain locally stored and are not directly shared.

This work presents a Federated Ensemble-Guided Pseudo-Labeling-based Semi-Supervised Learning (FEP-SSL) framework that integrates ensemble-driven pseudo-label generation with federated model aggregation. The framework is designed for binary PD classification, where each sample is labeled as 1 (PD) or 0 (healthy). It utilizes multiple boosting models to generate pseudo-labels through a consensus-based majority voting strategy, thereby reducing noise propagation and improving label reliability. The locally

trained models are subsequently aggregated using the Federated Averaging (FedAvg) [20] algorithm to construct a global model, while ensuring that raw patient data remain private and decentralized. By jointly leveraging SSL and FL within a unified framework, the proposed approach effectively utilizes limited labeled data, enables secure collaborative learning across distributed clients, and improves the robustness and generalization of PD classification. The framework is evaluated on two publicly accessible speech-based PD datasets. The key contributions of this study are summarized as follows:

- The proposed FEP-SSL framework integrates semi-supervised pseudo-label learning with federated optimization to enable privacy-preserving PD classification.
- A multi-model ensemble strategy based on majority voting is introduced to improve pseudo-label reliability and reduce noise propagation during training.
- An FL-based architecture incorporating hybrid SSL models is developed, where only model parameters are shared, facilitating effective learning under non-Independent and Identically Distributed (non-IID) data settings.
- Comprehensive ablation studies and comparative evaluations with state-of-the-art SSL methods are conducted, along with SHAP-based analysis to assess feature importance and enhance model interpretability.

The rest of the paper is organized as follows. Section 2 presents a review of related work. Section 3 describes the proposed FEP-SSL framework for PD classification. Section 4 details the experimental setup, reports the performance results, and provides a critical analysis of the findings. Finally, Section 5 concludes the paper.

2 LITERATURE REVIEW

Recent advances in ML have substantially improved automatic PD detection, particularly through the analysis of speech, handwriting, medical imaging, and other physiological signals. However, developing robust diagnostic models remains challenging due to limited amount of labeled clinical data and strict privacy constraints associated with patient information. As a result, recent research has increasingly focused on two complementary directions: SSL, which leverages both labeled and unlabeled data, and privacy-preserving distributed learning approaches such as FL.

The adoption of SSL has become as an essential strategy to address the scarcity of annotated medical data in PD detection. Early studies primarily focused on multi-view learning techniques to exploit complementary information from diverse feature representations. For instance, Zhang et al. [21] proposed a Semi-supervised Multi-view Learning Clustering (SMC) framework for PD screening using MRI data. Their approach leveraged multiple feature views to enhance clustering performance, demonstrating that integrating complementary representations can improve diagnostic accuracy. However, the method relies on manually defined feature consistency and may not generalize effectively to heterogeneous datasets. Similarly, Ammour et al. [22] explored SSL for analyzing handwriting patterns in PD patients, showing that motor impairments can be effectively captured even with limited labeled data. Despite its promising results, the approach is based on centralized datasets and may suffer from noise propagation when pseudo-labels are derived from low-confidence predictions. In another study, Bao et al. [23] extended the Semi-Supervised Competitive Learning (SSCL) framework for dysphonia voice classification, achieving an accuracy of 83.8% and a Receiver Operating Characteristic - Area Under Curve (ROC-AUC) of 93.3%. However, the method does not explicitly address class imbalance or pseudo-label uncertainty, which may limit its robustness in real-world scenarios.

More recent studies have explored advanced representation learning techniques for speech-based PD detection. Malekroodi et al. [24] employed pre-trained self-supervised Automatic Speech Recognition (ASR) models combined with supervised contrastive learning to enhance feature discriminability. Their approach attained an F1-score of 90% and a ROC-AUC of 89%, demonstrating the utility of self-supervised feature extraction in clinical speech analysis. In a related direction, Quatra et al. [25] proposed a Bilingual Dual-Head Deep Neural Network (BDH-DNN) architecture that leverages multilingual speech data for PD

detection. The model obtained 84.72% accuracy and a 69.03% F1-score on the EWA-DB dataset. Although the approach highlights the advantages of cross-lingual feature learning, it lacks explicit mechanisms for ensuring reliable pseudo-label selection.

Alongside advancements in SSL, ensuring data privacy has become a major concern in healthcare analytics due to stringent regulations on medical data sharing. In this context, FL offers a viable solution, enabling multiple institutions to jointly train models without sharing patient data. Several studies have explored FL-based approaches for PD detection and broader medical data analysis. For instance, Sarlas et al. [26] investigated an FL framework for speech-based PD detection, demonstrating the feasibility of distributed training in clinical settings. Their model gained 58.67% accuracy for the first client and 58.74% aggregated accuracy across clients. However, the approach does not address pseudo-label uncertainty or effectively utilize unlabeled data. Similarly, Danek et al. [27] applied federated Deep Learning (DL) to multi-omics data for PD analysis, highlighting the scalability of FL for large biomedical datasets. Nevertheless, their method primarily relies on fully labeled data and does not incorporate SSL strategies that could improve performance under limited labeled data conditions.

Other studies have focused on enhancing the robustness of FL systems under heterogeneous data conditions. Pang et al. [28] proposed a Tri-Factor Adaptive Federated Learning (TriAFL) framework for PD diagnosis using facial expression data, achieving an accuracy of 94.17%. However, the model does not address pseudo-label noise or effectively utilize unlabeled data sources. Similarly, Dipro et al. [29] introduced a federated DL framework for PD detection that achieved 97% accuracy, ensuring strong data privacy but relying exclusively on fully supervised data.

Beyond specific PD detection systems, several studies have investigated privacy-preserving learning techniques in broader healthcare contexts. Kaissis et al. [30] examined secure multi-party computation strategies for FL, demonstrating their effectiveness in mitigating model inversion and membership inference attacks. In addition, Ogundokun et al. [31] and Ali et al. [32] presented comprehensive surveys on FL applications in healthcare, highlighting key challenges such as communication overhead, data heterogeneity, and scalability. Furthermore, Qiu et al. [33] proposed a Federated Semi-Supervised Learning (FSSL) framework for medical image segmentation, incorporating pseudo-label denoising to reduce noise in predictions from unlabeled client data. The method yielded 88.89% accuracy and a 59.86% F1-score. However, the framework remains sensitive to pseudo-label quality, and its performance may degrade under highly non-IID data distributions across clients.

More recently, researchers have incorporated advanced privacy mechanisms and multimodal learning strategies within FL frameworks. Fares et al. [34] proposed a privacy-preserving FL architecture that integrates differential privacy and secure aggregation with a modified ResNet model. While the framework enhances privacy protection, it does not leverage unlabeled data to improve learning efficiency. Tanim et al. [35] introduced a Secure Federated Learning (SFL) approach based on homomorphic encryption to ensure data privacy under non-IID conditions. Although effective in protecting sensitive information, the method incurs significant computational overhead and increases training time. In another related study, Kaur et al. [36] developed a multimodal FL framework that combines multiple biomarkers for PD diagnosis. Despite achieving improved diagnostic performance, the model relies heavily on labeled multimodal data and does not exploit unlabeled data through SSL. Table 1 summarizes the key existing approaches for PD detection.

Despite significant progress, existing studies exhibit several critical limitations that restrict their applicability in real-world clinical settings. Most SSL-based approaches improve data efficiency but remain susceptible to pseudo-label noise, particularly in imbalanced and heterogeneous medical datasets. On the other hand, FL-based methods primarily emphasize privacy preservation while relying heavily on fully labeled data, thereby underutilizing the large volume of unlabeled data available in practical scenarios. In contrast, the proposed framework offers a unified solution to address these challenges. First, it incorporates an ensemble-based pseudo-labeling strategy to improve the robustness and reliability of pseudo-label selection compared to single-model SSL approaches. Second, the framework is designed to operate effectively without requiring large, fully labeled datasets, instead leveraging substantial unlabeled data to demonstrate robust performance, making it applicable in healthcare settings where labeled data are scarce and expensive. Third, the integration of multi-feature representations improves discriminative capability by capturing complementary information from heterogeneous speech features. Finally, the framework adopts a privacy-preserving distributed learning paradigm, method enabling collaborative learning across institutions

while preserving patient data privacy. In addition, FEP-SSL employs SHAP analysis to evaluate feature importance and provide global interpretability, offering clear insights into the factors driving PD classification, thereby improving the transparency and reliability of predictions.

3 Proposed FEP-SSL Framework for PD Classification

3.1 System Architecture

The FEP-SSL framework, illustrated in Fig. 1, integrates ensemble-based boosting models with SSL-based pseudo-labeling to effectively leverage unlabeled speech-derived features. A confidence-driven majority voting strategy is employed to improve the quality and reliability of pseudo-labels, while FL with the FedAvg algorithm facilitates collaborative model training across multiple clients. These components collectively establish a unified and scalable learning pipeline that enhances classification robustness and generalization performance across heterogeneous PD datasets.

The process begins with data acquisition, where two benchmark PD-related datasets are collected and integrated to enable robust and generalized model training. The dataset is processed and analyzed, including categorical encoding, class balancing using the Synthetic Minority Over-sampling Technique (SMOTE), normalization, generation of feature interaction terms, and a train-test split. Initially, the labeled subsets of speech-derived features are processed using ML models such as Extreme Gradient Boosting (XGB), Gradient Boosting (GB), and Light Gradient Boosting Machine (LGBM). These base learners are independently trained at each client to capture complementary decision patterns. Their predictions are subsequently combined using a majority voting strategy. To ensure the reliability of pseudo-labels, only predictions exceeding a predefined confidence threshold of 95% are retained and incorporated into the labeled dataset. Furthermore, stratified hold-out validation, also known as Monte Carlo cross-validation (MCCV), is applied iteratively to enhance robustness, reduce variance, and address class imbalance.

Subsequently, the framework then integrates FL through the FedAvg algorithm. Each client independently trains a local model using its own data, transmitting only model parameters, not raw data to a central server. The server aggregates these parameters to create an improved global model, which is redistributed to clients for further local refinement. This iterative process repeats until convergence, progressively enhancing the performance of the global model. The detailed procedure of the proposed framework is outlined in Algorithm 1.

Fig 1. Proposed FEP-SSL, a Federated Ensemble-Guided Pseudo-Labeling-based Semi-Supervised Learning framework for privacy-preserving PD classification.

3.2 Data Acquisition

Two acoustic feature-based datasets obtained from the UCI Machine Learning Repository are utilized in this study: one comprising 744 acoustic features [37], and another containing 48 features [38], referred to as Dataset 1 and Dataset 2, respectively.

3.2.1 Dataset 1

The dataset includes voice recordings from 188 individuals diagnosed with PD (107 males and 81 females), along with 64 healthy controls (23 males and 41 females). All voice samples were recorded at a sampling frequency of 44.1 kHz. Following clinical evaluation, participants were instructed to produce multiple repetitions of the sustained vowel sound /a/, and three audio recordings were collected from each participant [39].

Mel-Frequency Cepstral Coefficients (MFCCs) effectively model human auditory perception and are widely used in applications such as speaker recognition, automatic speech recognition [40], biomedical voice analysis [41], and PD diagnosis [42]. In the context of PD, MFCCs detect subtle variations in articulatory patterns, including tongue and lip motion. Another important approach for speech signal analysis is the

Table 1. Comparative summary of existing works.

Ref.	Methods	Dataset	Contribution	Performance	Limitations
Zhang et al. [21]	SMC	MRI Dataset	Screen PD using complementary MRI feature views	N/A	Relies on handcrafted features and limited generalization
Ammour et al. [22]	SSL	Handwriting Dataset	Characterize motor impairment in PD patients from handwriting	N/A	Centralized data is required, susceptible to pseudo-label noise
Bao et al. [23]	SSCL	Speech Dataset	Classify dysphonic speech signals for PD detection	Accuracy 83.8%, ROC-AUC 93.9%	Moderate performance, does not address class imbalance or pseudo-label uncertainty
Malekroodi et al. [24]	ASR	Speech Dataset	Improve speech representation for PD classification	F1-score 90.0%, ROC-AUC 89%,	Requires substantial labeled data, lacks distributed training support
Quatra et al. [25]	BDH-DNN	EWA-DB Speech Dataset	Improve cross-language PD detection using dual-head architecture	Accuracy 84.72%, F1-score 69.03%	Limited dataset size
Sarlas et al. [26]	FL framework	Speech Dataset	Enable privacy-preserving PD detection	Accuracy 58.74%	Does not utilize unlabeled data, lacks pseudo-label confidence mechanisms
Danek et al. [27]	Federated DL	Multi-omics Dataset	Evaluate FL for PD analysis	ROC-AUC 87.6%	Performance is sensitive to sample dispersion across clients
Pang et al. [28]	TriAFL	PDFE-V2 dataset	Collaborative analysis of PD facial expressions using FL	Accuracy 94.17%	Does not address pseudo-label noise or unlabeled data utilization
Dipro et al. [29]	Federated DL	Speech Dataset	Privacy-preserving PD detection with distributed medical data	Accuracy 97%	Fully supervised, cannot handle label scarcity
Kaissis et al. [30]	Secure multi-party FL	Medical Data	Mitigate model inversion & membership inference attacks	N/A	Computationally complex; not tested on PD-specific datasets
Qiu et al. [33]	FSSL	RSNA ICH, ISIC 2018	Combine SSL with FL for better generalization	Accuracy 88.89%, F1-score 59.86%	Performance varies under non-IID client data
Tanim et al. [35]	SFL	Bio-StampRC21	Protect sensitive PD data under non-IID distributed environments	Accuracy 64.40%	Added encryption overhead; limited external benchmarking
Kaur et al. [36]	Multimodal FL	Multimodal Dataset	Integrate multiple PD biomarkers for improved diagnosis	Accuracy 85%	Heavily dependent on labeled multimodal data; lacks SSL capability

Table 2. Overview of feature categories extracted from Dataset 1.

Category	Measures	Features
Baseline features	Jitter variants	5
	Shimmer variants	6
	Fundamental frequency parameters	5
	Harmonicity parameters	2
	Recurrence Period Density Entropy	1
	Detrended Fluctuation Analysis	1
	Pitch Period Entropy	1
	Intensity parameters	5
	Formant frequencies	4
Tunable Q-factor wavelet	Bandwidth	4
	TQWT features related to F_0	432
Wavelet transform based	WT features related to F_0	182
	Glottis quotient	3
	Glottal to noise ratio	6
Vocal fold features	Vocal fold excitation ratio	7
	Empirical mode decomposition	6
Mel-frequency cepstral coefficient	MFCCs	84

Wavelet Transform (WT), which is particularly effective for analyzing signals with localized or transient variations. Several studies have utilized WT-derived features, especially those related to the fundamental frequency (F_0), to detect PD, as they reflect irregularities in vocal patterns [43]. In addition, the Tunable Q-Factor Wavelet Transform (TQWT) is employed as a feature extraction technique. It is governed by three parameters: Q (quality factor), r (redundancy), and J (number of decomposition levels), which can be adjusted to adapt the wavelet representation to the characteristics of the signal. In addition to these features, several attributes related to vocal fold vibration are included to assess the impact of noise on vocal behavior. Detailed information on the extracted feature types is presented in Table 2.

3.2.2 Dataset 2

The PD speech dataset [44] comprises acoustic features extracted from sustained phonation of the vowel /a/, including voice recordings from 80 subjects, with 40 diagnosed with PD and 40 serving as healthy controls. The dataset is multivariate in nature and falls within the domain of health and medical studies.

The acoustic features capture key characteristics of PD-affected speech, including vocal fold instability (jitter and shimmer), degradation in voice quality, and nonlinear phonatory irregularities such as RPDE, DFA, PPE, and GNE. In addition, MFCCs and their temporal delta coefficients represent spectral and dynamic variations in speech, enabling effective discrimination between PD patients and healthy controls. A comprehensive summary of these features, including their physiological and signal-processing aspects, is presented in Table 3. Each instance corresponds to one of three recordings per subject; therefore, samples from the same individual cannot be considered statistically independent. Although the dataset contains 240 samples, these correspond to 80 unique subjects [45]. As a result, applying traditional ML algorithms under the assumption of independent and identically distributed (IID) samples may lead to biased performance estimates. Unlike conventional repeated measurements, the replication scheme in this dataset involves feature sets derived from multiple recordings of the same sustained phonation task. While these features are expected to be similar, slight variations arise due to biological differences and limitations in recording and feature extraction processes. Consequently, samples from the same subject exhibit higher similarity compared to those from different subjects [46]. To address these dependencies, subject-aware evaluation strategies or learning approaches designed for dependent data should be employed, as suggested in prior studies [45]. This

Table 3. Overview of feature categories extracted from Dataset 2.

Category	Measures	Features
Metadata	Recording, Gender	3
Jitter	Jitter_rel, Jitter_abs, Jitter_RAP, Jitter_PPQ	4
Shimmer	Shim_loc, Shim_dB, Shim_APQ3, Shim_APQ5, Shi_APQ11	5
Noise	HNR05, HNR15, HNR25, HNR35, HNR38	5
Nonlinear	RPDE, DFA, PPE, GNE	4
MFCC	MFCC0-MFCC12	13
Temporal	Delta0-Delta12	13

ensures a more reliable and unbiased assessment of model performance in PD detection tasks.

3.3 Data Preprocessing and Feature Representation

The datasets comprise both categorical and numerical features, resulting in a complex feature structure that poses challenges for accurate classification. This complexity necessitates careful preprocessing and effective feature management. To address this, a multi-stage preprocessing pipeline is implemented to prepare the data for ML and FL models. Initially, label encoding [47] is applied to convert categorical binary variables, such as gender, into numerical representations, ensuring compatibility with ML algorithms. Another critical aspect is the presence of class imbalance, a common issue where certain classes are underrepresented. To mitigate this, the Synthetic Minority Over-sampling Technique (SMOTE) [48] is employed to generate synthetic samples for the minority class through linear interpolation between neighboring instances. This process enhances class distribution balance, promoting balanced model performance across classes and minimizing skew toward the larger class.

After preprocessing, the datasets are split into feature sets and target sets. To promote unbiased training and evaluation of the model, the data are partitioned into multiple subsets. Specifically, a hold-out test set comprising 20% of the data is reserved solely for the final performance assessment. The remaining 80% of the data, referred to as the working pool, is further divided into an initial labeled set (10%), a validation set (10%), and an unlabeled pool (80%). This partitioning strategy simulates low-label scenarios commonly encountered in medical applications and facilitates iterative SSL.

3.4 Base Learners: Three-Model Ensemble

Under the combined multiple feature types setting, three ensemble-based algorithms are developed and trained on equal portions of the aggregated feature sets. The framework integrates ensemble-guided pseudo-labeling with federated optimization to enable robust model training while preserving data privacy. A supervised classification setting is considered, where a small set of labeled samples is available, accompanied by a large amount of unlabeled data. The labeled dataset ($\mathcal{L}$) is defined as in Eq. 1.

$$\mathcal{L} = \{(x_i, y_i)\}_{i=1}^{N_{\mathcal{L}}} \quad (1)$$

where x_i denotes the input feature vector of a labeled sample, y_i represents the corresponding class label, $N_{\mathcal{L}}$ is the total number of labeled samples, and i is the index of the labeled data samples. The unlabeled dataset ($\mathcal{U}$) is defined in Eq. 2.

$$\mathcal{U} = \{x_j\}_{j=1}^{N_{\mathcal{U}}} \quad (2)$$

where x_j denotes the feature vector of an unlabeled sample, $N_{\mathcal{U}}$ represents the total quantity of unlabeled samples, and j is the index of the unlabeled data samples. The validation set ($\mathcal{V}$) and the test set ($\mathcal{W}$) are defined in Eq. 3 and Eq. 4, respectively.

$$\mathcal{V} = \{(x_p)\}_{p=1}^{N_V} \quad (3)$$

$$\mathcal{W} = \{(x_q)\}_{q=1}^{N_W} \quad (4)$$

where x_p represents the feature vector of a validation sample, N_V denotes the total number of validation samples, and p is the index of the validation data. Similarly, x_q denotes the feature vector of a test sample, N_W represents the number of test samples, and q is the index of the test data. To promote model diversity, the labeled data are divided into three equal feature subsets, denoted as F_1 , F_2 , and F_3 . Each subset is independently processed by ensemble classifiers. The three base learners XGB, GB, and LGBM are denoted as M_1 , M_2 , and M_3 , respectively.

XGB [49] enhances gradient boosting with regularization, though improper tuning may lead to overfitting. It accelerates the boosting process through parallel computation and optimized tree-learning algorithms, thereby improving efficiency and scalability on large datasets. Due to its strong balance between predictive accuracy and computational performance, XGB has become one of the most widely used boosting algorithms. The objective of XGB is to minimize a loss function (L), as defined in Eq. 5.

$$L = \sum_{n=1}^{N_x} \ell(y_n, \hat{y}_n) + \Omega(f) \quad (5)$$

where $\ell(y_n, \hat{y}_n)$ denotes the loss function that measures the discrepancy between the true label y_n and the predicted label $\hat{y}_n$, $\Omega(f)$ represents the regularization term, and N_x corresponds to the total size of the training samples.

GB [50] is an iterative ensemble technique in which successive weak learners are trained to correct the errors of their predecessors. Typically implemented using decision trees as base learners, GB minimizes a specified loss function through gradient-based optimization. In practice, directly estimating the model parameters for a given loss function or custom base learner can be challenging. To address this, the learning process is guided by the negative gradient $\{g_t(x)\}$, computed from the observed data as defined in Eq. 6.

$$g_t(x) = \mathbb{E}_y \left[\frac{\partial \hat{y}(y, f(x))}{\partial f(x)} \middle| x \right], f(x) = f^{t-1}(x) \quad (6)$$

where $\frac{\partial \hat{y}(y, f(x))}{\partial f(x)}$ denotes the derivative of the loss function with respect to the model prediction, $f(x)$ represents the current prediction, and $f^{t-1}(x)$ corresponds to the prediction from the previous iteration.

LGBM [51] is a gradient boosting framework designed to improve training efficiency in terms of speed and memory usage by employing histogram-based representation and a leaf-wise tree growth strategy. It is well-suited for handling large-scale datasets while maintaining high predictive accuracy. The leaf-wise growth approach allows the model to prioritize splits that yield the maximum reduction in loss, often resulting in better performance compared to traditional level-wise tree construction methods [52]. The gain from a split is computed as defined in Eq. 7.

$$\text{Gain} = \frac{1}{2} \left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right) - \gamma \quad (7)$$

Where G represents the total gradients of all instances in the current node, and H denotes the sum of the corresponding Hessians. Similarly, G_L and H_L refer to the gradients and Hessians of the left child node, while G_R and H_R correspond to those of the right child node. The parameters λ and γ act as regularization terms to regulate model complexity.

3.5 FEP-SSL Model

The proposed FEP-SSL framework generates pseudo-labels for unlabeled samples using a majority voting strategy with a confidence threshold of 0.95, assigning labels only when at least two base models agree to ensure low-noise labeling. Each client then trains a local hybrid model on its own data, and only the model parameters are shared and aggregated using FedAvg, thereby preserving data privacy.

Algorithm 1: Federated Ensemble-Guided Pseudo-Labeling-based Semi-Supervised Learning (FEP-SSL)

Require: Labeled dataset $\mathcal{L}$, unlabeled data $\mathcal{U}$, Number of clients K , learning rate η , communication rounds R , threshold τ , ensemble model M_{best} from Algorithm 2

Ensure: Optimized global hybrid model M_{global}

- 1: Train three base learners on disjoint feature subsets
- 2: Generate pseudo-labels for $\mathcal{U}$ using majority voting
- 3: Expanded labeled dataset and retrain ensemble $M_{\text{best}} \leftarrow \{M_1, M_2, M_3\}$ on updated dataset
- 4: Initialize global parameters $\Theta^{(0)}$ using M_{best}
- 5: Distribute $\Theta^{(0)}$ to all clients $C_{k=1}^K$
- 6: **for all** clients C_k **do**
- 7: Initialize local models $\theta_k^{(m,0)} = \theta_{\text{global}}^{(m,0)}$
- 8: **end for**
- 9: **for** $r = 1$ **to** R **do**
- 10: **for all** clients C_k **in parallel do**
- 11: **for** $m = 1$ **to** 3 **do**
- 12: $\theta_k^{(m,r)} \leftarrow \theta_k^{(m,r-1)} - \eta \nabla_{\theta_k^{(m,r-1)}} L_k^{(m)}(\theta_k^{(m,r-1)})$
- 13: **end for**
- 14: **end for**
- 15: **for** $m = 1$ **to** 3 **do**
- 16: $\theta_{\text{global}}^{(m,r)} \leftarrow \sum_{k=1}^K \frac{n_k}{N} \theta_k^{(m,r)}$
- 17: **end for**
- 18: $\Theta^{(r)} \leftarrow \{\theta_{\text{global}}^{(1,r)}, \theta_{\text{global}}^{(2,r)}, \theta_{\text{global}}^{(3,r)}\}$
- 19: Distribute $\Theta^{(r)}$ to all clients
- 20: **end for**
- 21: $M_{\text{global}} \leftarrow \text{Model}(\Theta^{(r)})$
- 22: Compute feature importance using SHAP

$$\phi \leftarrow \text{SHAP}(M_{\text{global}}, \mathcal{L})$$

22: **return** M_{global}, ϕ

3.5.1 Ensemble-Guided Pseudo-Labeling

Following the initial supervised training phase on the available labeled data, the pseudo-labeling process is initiated. The base learners are first trained independently to capture complementary decision patterns from the labeled subset, and are then integrated into a unified ensemble framework. The trained ensemble is applied to the unlabeled data, where each base learner generates class probability estimates. A pseudo-label is assigned based on majority voting with a predefined confidence threshold. This strategy effectively reduces ambiguous and noisy predictions, thereby improving the reliability of pseudo-labels.

Most SSL approaches produce pseudo-labels based on the predictions of a single model, without consulting multiple models for consensus. While this enables the use of unlabeled data, it also makes the process highly susceptible to noisy pseudo-labels. Such noise can lead to error propagation, frequent retraining, and greater dependence on manual annotations, thereby increasing both training and labeling costs [53, 54]. These issues are particularly pronounced in large, domain-sensitive applications, where labeling errors can significantly undermine model stability and the effectiveness of systematic learning.

To mitigate these issues, the framework employs an SSL model with majority voting, which has been demonstrated to produce more reliable pseudo-labels while lowering overall training costs [55]. In each iteration, pseudo-labels are assigned to unlabeled samples only when a majority of the models agree on the same label. This majority voting mechanism effectively filters out ambiguous predictions, ensuring that only

Fig 2. Majority voting-based pseudo-label generation using multiple classifiers.

trustworthy pseudo-labels are incorporated into the expanded labeled set. The pseudo-label assignment procedure is formally defined in Eq. 8.

$$\hat{y}_{\text{pseudo}} = \begin{cases} c, & \sum_{m=1}^3 \mathbb{I}(y_{(m)} = c) \geq 2 \\ \text{Discard}, & \text{Otherwise,} \end{cases} \quad (8)$$

where $\hat{y}_{\text{pseudo}}$ represents the final pseudo-label assigned to an unlabeled sample, $m \in \{1, 2, 3\}$ indexes the base models in the ensemble, $y_{(m)}$ denotes the class prediction produced by the m -th model, and $\mathbb{I}(\cdot)$ is the indicator function. To further enhance reliability and mitigate labeling noise, the framework employs a hybrid ensemble strategy that combines predictions from multiple models. As shown in Fig. 2, the initially labeled training set is used to collectively train a hybrid ensemble denoted as $M_{\text{hybrid}} = \{M_1, M_2, M_3\}$. Each model independently processes the unlabeled data, generating predictions y_1 , y_2 , and y_3 . Rather than assigning pseudo-labels based on a single model's output, the framework incorporates a majority voting module. When a sample meets the confidence criterion, it is assigned a pseudo-label and added to the training set for subsequent iterations. By leveraging inter-model consensus and confidence filtering, the framework significantly reduces labeling noise at a manageable computational cost. Only high-confidence, consensus-based samples are retrained, minimizing redundancies and preventing the propagation of errors.

All newly accepted pseudo-labeled samples $\mathcal{P}$ are added to the labeled dataset $\mathcal{L}$. This incorporation of accepted pseudo-labeled samples into the labeled set is formally defined in Eq. 9.

$$\mathcal{L}^{(t+1)} = \mathcal{L}^{(t)} \cup \{(x_j, \hat{y}_{\text{pseudo}})\} \quad (9)$$

The pseudo-labeled samples that meet the acceptance criteria are incorporated into the labeled dataset and simultaneously removed from the unlabeled set. Subsequently, each base learner is retrained on this expanded labeled set by minimizing the regularized cross-entropy loss defined in Eq. 10.

$$\theta_m^* = \arg \min_{\theta_m} \sum_{(x,y) \in \mathcal{L}} \mathcal{L}_{\text{CE}}(y, M_m(x; \theta_m)) + \lambda \Omega(\theta_m) \quad (10)$$

where $M_m(x; \theta_m)$ denotes the output of the m -th base learner with parameters θ_m , θ_m^* represents the optimal parameters of the m -th model after retraining, $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss function, $\Omega(\theta_m)$ is the regularization term, and λ is the regularization coefficient controlling model complexity. The optimal hybrid model is then selected according to validation performance, as defined in Eq. 11.

$$M_{\text{best}} = \arg \max_{\mathcal{M}^{(t)}} \text{Eval}(\mathcal{M}^{(t)} \mathcal{V}) \quad (11)$$

where $M^{(t)}$ denotes the hybrid ensemble model obtained at iteration t of the pseudo-labeling process. To ensure model stability and reduce overfitting, the framework retains the hybrid model that performs best on the validation set. The model is evaluated iteratively throughout training and the current validation score is compared with the previously recorded best score. When $\text{score}_{\text{current}} > \text{score}_{\text{best}}$, the corresponding model parameters are updated and stored as the best model. The pseudo-labeling process terminates when either of the following conditions is met: (a) no new pseudo-labeled samples are added in an iteration, or (b) no improvement in validation performance is observed over two consecutive iterations.

The proposed ensemble-guided pseudo-labeling mechanism integrates majority voting with confidence thresholding to generate reliable pseudo-labels. This approach enhances labeling quality, reduces annotation costs, and lowers computational overhead. By accepting only high-confidence pseudo-labels, the framework minimizes reliance on manual annotation—particularly costly and time-consuming in healthcare and biomedical applications. The consensus mechanism also filters out low-confidence predictions, preventing contamination of the training set and avoiding unnecessary retraining. Consequently, the labeled dataset is progressively expanded, improving training stability and accelerating convergence, as outlined in Algorithm 2.

Algorithm 2: Ensemble-Guided Pseudo-Labeling

Require: Labeled dataset $\mathcal{L}$, unlabeled dataset $\mathcal{U}$, validation dataset $\mathcal{V}$, Test dataset $\mathcal{W}$, threshold τ

Ensure: Optimized ensemble model M_{best} , maximum pseudo-labeling iterations T

```
1: Partition the feature space of  $\mathcal{L}$  into three equal-sized, pairwise disjoint subsets  $F_1$ ,  $F_2$ , and  $F_3$ .
2: Train base learners
3:    $M_1 \leftarrow \text{XGB}(\mathcal{L}[F_1])$ 
4:    $M_2 \leftarrow \text{GB}(\mathcal{L}[F_2])$ 
5:    $M_3 \leftarrow \text{LGBM}(\mathcal{L}[F_3])$ 
6:  $M_{\text{hybrid}} \leftarrow \{M_1, M_2, M_3\}$ 
7:  $\text{score}_{\text{best}} \leftarrow \text{Evaluate}(M_{\text{hybrid}}, \mathcal{V})$ 
8:  $M_{\text{best}} \leftarrow M_{\text{hybrid}}$ 
9: for  $t = 1$  to  $n$  do
10:    $\mathcal{P} \leftarrow \emptyset$ 
11:   for all  $x_j \in \mathcal{U}$  do
12:      $y_1 \leftarrow M_1(x_j)$ 
13:      $y_2 \leftarrow M_2(x_j)$ 
14:      $y_3 \leftarrow M_3(x_j)$ 
15:      $(\hat{y}_{\text{pseudo}}, \text{conf}) \leftarrow \text{MajorityVote}(y_1, y_2, y_3)$ 
16:     if  $\text{conf} \geq \tau$  then
17:        $\mathcal{P} \leftarrow \mathcal{P} \cup \{(x_j, \hat{y}_{\text{pseudo}})\}$ 
18:     end if
19:   end for
20:   if  $|\mathcal{P}| = 0$  then
21:     break
22:   end if
23:    $\mathcal{L} \leftarrow \mathcal{L} \cup \mathcal{P}$ 
24:    $\mathcal{U} \leftarrow \mathcal{U} \setminus \{x_j \mid (x_j, y_j) \in \mathcal{P}\}$ 
25:   Retrain  $M_{\text{hybrid}}$  on updated  $\mathcal{L}$ 
26:    $\text{score}_{\text{current}} \leftarrow \text{Evaluate}(M_{\text{hybrid}}, \mathcal{V})$ 
27:   if  $\text{score}_{\text{current}} > \text{score}_{\text{best}}$  then
28:      $\text{score}_{\text{best}} \leftarrow \text{score}_{\text{current}}$ 
29:      $M_{\text{best}} \leftarrow M_{\text{hybrid}}$ 
30:   end if
31:   if  $t \geq 2$  and no improvement in last two iterations then
32:     break
33:   end if
34: end for
35: return  $M_{\text{best}}$ 
```

341

3.5.2 Federated Learning with Hybrid Local Models

342

Centralized learning in healthcare carries substantial privacy and regulatory risks due to the transfer and storage of sensitive patient data. To address these concerns, the proposed framework, integrated with FedAvg [20], keeps patient data locally at each client, exchanging only model parameters with the central server, thereby preserving data privacy. To emulate realistic clinical data heterogeneity, the local datasets across clients are partitioned using a Dirichlet distribution with concentration parameter $\beta = 0.8$, which introduces non-IID characteristics by assigning varying class proportions to each client. Such a setting reflects real-world scenarios where hospitals encounter diverse patient populations with different disease distributions. Conventional FL approaches often face challenges under non-IID data distributions, which can lead to unstable convergence and reduced generalization performance. In contrast, the proposed framework is explicitly designed to maintain stable learning under such heterogeneous conditions [56]. Each client employs three complementary sub-models to capture diverse data characteristics, while a confidence-aware SSL

343
344
345
346
347
348
349
350
351
352
353

pseudo-labeling method is applied locally. The global model is updated periodically through iterative communication rounds.

The FL setup is evaluated on speech data from individuals exhibiting diverse acoustic characteristics and varying disease severity, reflecting real-world clinical scenarios. This heterogeneity is additionally reinforced through the use of a Dirichlet distribution-based partitioning strategy, ensuring that each client possesses distinct yet representative data distributions. By integrating client-specific hybrid local models, confidence-aware pseudo-labeling, and federated aggregation, the framework effectively handles this heterogeneity, achieving stable convergence and robust global performance. The system operates in a collaborative learning environment comprising a central server and a fixed set of K participating clients, denoted as $\{C_1, C_2, \dots, C_K\}$. Each client C_k maintains a private local dataset $\mathcal{D}_k$, which remains on-device and is never transmitted to the central server. The union of all client datasets constitutes the global dataset $\mathcal{D}$. The size of client k 's local dataset is denoted by $n_k = |\mathcal{D}_k|$, representing the number of samples stored at that client. Prior to the start of FL, the global parameters are initialized with the best hybrid model M_{best} obtained from the pseudo-labeling phase, as defined in Eq. 12.

$$\Theta(0) = \{\theta_{\text{global}}^{(1,0)}, \theta_{\text{global}}^{(2,0)}, \theta_{\text{global}}^{(3,0)}\} \quad (12)$$

Each communication round comprises a coordinated between the server and the involved clients. The local update for each model type m as defined in Eq. 13.

Each client initializes its local hybrid model by adopting the current global parameters. The learning process proceeds iteratively across a sequence of communication rounds indexed by $r = 1, 2, \dots, R$. Each round involves coordinated interactions between the central server and all participating clients. The local update for each model type m is defined in Eq. 13.

$$\theta_k^{(m,r)} = \theta_k^{(m,r-1)} - \eta \nabla_{\theta_k^{(m,r-1)}} L_k^{(m)}(\theta_k^{(m,r-1)}) \quad (13)$$

given:

$$L_k^{(m)}(\theta_k^{(m,r-1)}) = \frac{1}{n_k} \sum_{(x,y) \in \mathcal{D}_k} \ell^{(m)}\left(y, M_m(x^{(m)}, \theta_k^{(m,r-1)})\right)$$

where $\theta_k^{(m,r)}$ and $\theta_k^{(m,r-1)}$ represent the parameters of the m -th sub-model at client k after the current and previous communication rounds, respectively. The term $\nabla_{\theta_k^{(m,r-1)}}$ denotes the gradient with respect to $\theta_k^{(m,r-1)}$, and $L_k^{(m)}(\theta_k^{(m,r-1)})$ is the corresponding local loss function. The global update after round r is defined in Eq. 14.

$$\Theta^{(r)} = \left\{ \theta_{\text{global}}^{(m,r)} \right\}_{m=1}^3 \quad (14)$$

given:

$$\theta_{\text{global}}^{(m,r)} = \sum_{k=1}^K \frac{n_k}{N} \theta_k^{(m,r)},$$

Let $\Theta^{(r)}$ represent the global model parameter set obtained after the r -th communication round in the FL process. The final global hybrid model is then formed by aggregating these parameters, as defined in Eq. 15.

$$M_{\text{global}} = \text{Model}(\Theta^{(r)}) \quad (15)$$

Overall, the proposed model improves learning performance, addresses data heterogeneity, lowers annotation costs, and facilitates privacy-preserving PD detection.

3.6 Model Interpretability

Shapley Additive Explanations (SHAP) is a model-agnostic interpretability method that explains ML predictions by assigning Shapley values to individual features, reflecting their contributions based on principles from cooperative game theory [57]. By quantifying the magnitude of feature contributions, SHAP

evaluates both the significance of features and their interactions across the dataset. In this study, SHAP is applied to test samples to derive global model interpretability, enabling a holistic understanding of how features influence the model’s predictions beyond individual instances. This approach enhances the transparency of model decisions and offers detailed insights into the role of each feature. The Shapley value for a feature b is computed as defined in Eq. 16.

$$\phi_b = \sum_{S \subseteq N \setminus \{b\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{b\}) - v(S)] \quad (16)$$

where ϕ_j is the Shapley value associated with feature b , S is a feature subset excluding b , N represents the set of all features, and $v(S)$ denotes the model output corresponding to the feature set S . The difference $v(S \cup \{b\}) - v(S)$ captures the marginal contribution of feature b to the subset S , while the factorial-based coefficient ensures that contributions are fairly averaged across all possible feature orderings. Intuitively, Shapley values quantify the average marginal contribution of a feature over all subsets, providing a theoretically grounded measure of feature importance. Examining these values allows identification of the most influential features and their interactions, thereby enhancing model interpretability.

3.7 Cross-validation protocol

Each dataset in this study contains three voice recordings per individual, requiring strict subject-level separation between train test sets to avoid data leakage. The proposed models are evaluated using Monte Carlo Cross-Validation (MCCV) over five rounds, ensuring that recordings from the same individual do not appear in both phases. MCCV provides a more realistic and unbiased assessment of model performance by maintaining subject-level separation while repeatedly sampling stratified training and testing splits, resulting in more reliable classification outcomes. This validation strategy reduces performance variability, mitigates class imbalance, and enhances model robustness and generalizability. In contrast, many existing studies employ leave-one-out cross-validation, which can yield biased estimates when multiple samples per subject exist, as recordings from the same individual may influence both training and testing sets.

4 Result and Discussion

This section provides a comprehensive overview of the experimental results, covering the experimental protocol and evaluation setup, SSL progression, ablation studies, comparison with existing SSL algorithms, and the framework’s interpretability using SHAP.

4.1 Experimental setup

The experiments are performed using Python in the Google Colaboratory (Colab) environment with an NVIDIA T4 GPU. To ensure fair and consistent comparisons, all classifiers are trained and evaluated using their default hyperparameters. Data integrity is maintained, and leakage is prevented by assigning each participant’s data to a single subset—labeled, unlabeled, validation, or testing—without overlap. Furthermore, an ensemble-guided pseudo-labeling strategy is applied to effectively utilize unlabeled data within the SSL framework. To simulate realistic client heterogeneity, the data across participants are partitioned using a Dirichlet distribution with concentration parameter $\beta = 0.8$, reflecting the variable class distributions observed in real-world clinical settings. For clarity, the study defines four distinct model stages as follows:

Initial Model: The model is trained solely on the labeled dataset.

Retrained Model: The model is trained on the dataset after applying pseudo-labeling.

Final Model: The retrained model is further refined using majority voting, leveraging both the expanded labeled dataset and pseudo-labeled samples to enhance performance.

Global Model: The model is obtained by aggregating pseudo-labeled datasets from all participating clients using the FedAvg algorithm.

The classification task is framed as a binary problem, where 1 represents a PD patient and 0 represents a

healthy individual. To thoroughly assess predictive performance, multiple evaluation metrics are employed, including Accuracy, Macro-average Precision (maP), Macro-average Recall (maR), Macro-average F1 (maF1), ROC-AUC, Matthews Correlation Coefficient (MCC), and Cohen’s kappa. These metrics are calculated based on True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN). This thorough evaluation offers a balanced measure of model performance, particularly under class imbalance, by effectively accounting for both positive and negative prediction outcomes. The key parameters setting are summarized in Table 4.

Table 4. Hyperparameter settings in the proposed framework.

Parameter	Value
Random seed	42
Learning rate	0.1
Confidence threshold (τ)	≥ 0.95
N estimators	120
Max depth	6
Subsample	0.8
Eval metric	logloss
Stratified sampling	Yes
Pseudo-labeling iterations	20
Number of clients	10
Local epochs per client	5
Dirichlet concentration (non-IID)	0.8
Aggregation method	FedAvg

Accuracy: Accuracy quantifies the performance of a classification model by representing the percentage of correct predictions (TP and TN) out of all predictions, as defined in Eq. 17.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

Macro-average Precision (maP): In multi-class classification, maP computes the precision for each class and then averages these values across all classes N , as defined in Eq. 18

$$\text{maP} = \frac{1}{N} \sum_{c=1}^N \frac{TP_c}{TP_c + FP_c} \quad (18)$$

Macro-average Recall (maR): maR is the unweighted average of per-class recall, providing an objective measure of model performance, particularly in imbalanced datasets, as defined in Eq. 19.

$$\text{maR} = \frac{1}{N} \sum_{c=1}^N \frac{TP_c}{TP_c + FN_c} \quad (19)$$

Macro-average F1 (maF1): The maF1 is used for multi-class classification and is computed by calculating the F1 score for each class individually and then taking the arithmetic mean across all classes, as defined in Eq. 20.

$$\text{maF1} = \frac{1}{N} \sum_{c=1}^N \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (20)$$

ROC-AUC: The ROC-AUC evaluates the model’s capability to differentiate classes by assessing the trade-off between True Positive Rate (TPR) and False Positive Rate (FPR) across thresholds, as defined in Eq. 21.

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx \quad (21)$$

Matthews Correlation Coefficient (MCC): MCC incorporates all four components—TP, TN, FP, and FN—and is especially effective for assessing performance on imbalanced datasets, as defined in Eq. 22.

$$\text{MCC} = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (22)$$

Cohen’s Kappa: The Kappa coefficient quantifies the agreement between predicted and true labels, while accounting for the level of agreement expected by random chance, as defined in Eq. 23.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (23)$$

where p_o represents the observed agreement and p_e denotes the agreement expected by chance.

4.2 SSL Progression

The effectiveness of the proposed ensemble-guided pseudo-labeling strategy is assessed by tracking the iteration-wise performance progression of the model on the complete feature set. This evaluation demonstrates that classification quality can be enhanced through iterative pseudo-label refinement within the same model family, without altering the underlying architecture. Fig. 3 presents the performance history for Dataset 1, showing maF1 and MCC across SSL training iterations. Both metrics exhibit a steady upward trend, with substantial performance gains observed as iterative pseudo-labels are incorporated. The peak performance occurs at iteration 7, achieving 95.55% maF1 and 91.48% MCC, reflecting optimal class-balanced discrimination and strong correlation between predicted and true labels. Although minor fluctuations appear in subsequent iterations, the overall trend confirms that the SSL framework effectively enhances generalization while avoiding overfitting.

Fig 3. Performance progression of the proposed framework across iterations on Dataset 1, measured using maF1 and MCC.

Fig. 4 illustrates the evolution of maF1 and MCC across SSL iterations for Dataset 2 using the complete feature set. The iterative pseudo-label refinement process gradually improves performance, reaching approximately 71.31% maF1 and 50.59% MCC by iteration 5. A temporary dip occurs at iteration 7, where maF1 and MCC decrease to 68.70% and 46.20%, respectively, indicating minor instability during pseudo-label updates. Nevertheless, the model stabilizes in subsequent iterations and attains its peak performance at iteration 12, achieving 85.01% maF1 and 73.12% MCC, representing the highest overall classification performance.

Fig 4. Performance progression of the proposed framework across iterations on Dataset 2, measured using maF1 and MCC.

4.3 Model Comparison Across Datasets

A concise summary of the proposed framework’s effectiveness is provided through performance comparisons across all evaluated datasets, emphasizing its generalization capability and consistency.

Table 5. Comparison of Initial, Retrained, Final, and Global Models’ performance on Dataset 1.

Model	Accuracy (%)	maF1 (%)	MCC (%)	Kappa (%)
Initial Model	83.89 ± 5.34	70.96 ± 11.35	46.14 ± 22.58	43.26 ± 21.51
Retrained Model	93.63 ± 3.22	87.22 ± 6.50	75.67 ± 12.92	74.54 ± 12.94
Final Model	97.18 ± 0.81	94.69 ± 1.63	89.94 ± 2.96	89.41 ± 3.25
Global Model	97.14	97.14	97.14	96.43

As presented in Table 5, the Initial Model attains an accuracy of 83.89% (± 5.34), whereas the Final Model substantially improves performance, achieving an accuracy of 97.18% (± 0.81) through iterative training. While the Final Model reaches an MCC of 89.94% (± 2.96), the Global Model demonstrates superior overall consistency, attaining a peak maF1 of 97.14%, an MCC of 97.14%, and a maximum Kappa value of 96.43%. In comparison, Sakar et al. [58] reported a maximum accuracy of 89%, with an F1-score of 84% and MCC of 59% using feature subset selection. The proposed framework outperforms this approach, achieving an accuracy of 97.14% and a maF1 of 97.14%.

Table 6. Comparison of Initial, Retrained, Final, and Global Models performance on Dataset 2.

Model	Accuracy (%)	maF1 (%)	MCC (%)	Kappa (%)
Initial Model	80.62 $\pm$ 5.59	68.82 $\pm$ 12.54	40.88 $\pm$ 21.71	39.34 $\pm$ 22.88
Retrained Model	86.83 $\pm$ 2.78	74.02 $\pm$ 7.66	52.53 $\pm$ 12.99	49.07 $\pm$ 14.44
Final Model	92.24 $\pm$ 4.29	85.00 $\pm$ 8.20	73.12 $\pm$ 14.70	70.46 $\pm$ 16.08
Global Model	95.50	95.50	95.30	93.05

Similarly, as shown in Table 6, the performance on Dataset 2 improves steadily across the training stages. The Initial Model gains an accuracy of 80.62% (± 5.59), while the Retrained Model increases accuracy to 86.83% (± 2.78). The Final Model further enhances discriminative capability, reaching 92.24% (± 4.29) accuracy, with notable gains in maF1 (85.00% ± 8.20) and MCC (73.12% ± 4.70). The Global Model attains the best overall performance, achieving 95.50% accuracy and maF1, along with the highest MCC of 95.30% and Kappa of 93.05%. In comparison, Naranjo et al. [45] reported maximum accuracy and precision of 70% and 81%, respectively, using selected feature subsets, highlighting the limitations of handcrafted feature selection and subject-level Bayesian assumptions, which reduce scalability for larger datasets. These results demonstrate that the proposed framework effectively enhances model performance through iterative pseudo-labeling and retraining.

4.4 Final Model Performance

To further evaluate the generalization capability of the Final Model, its performance is tested on independent validation and test datasets. The model demonstrates high accuracy and robust agreement metrics, confirming its reliability and stability on previously unseen samples.

Table 7. Final Model validation and test performance on Dataset 1 and Dataset 2.

Dataset	Model	Accuracy (%)	maF1 (%)	MCC (%)	Kappa (%)
Dataset 1	Final Model (Validation)	96.87	96.87	96.87	95.50
	Final Model (Test)	96.52	96.52	96.52	95.58
Dataset 2	Final Model (Validation)	93.40	93.40	93.40	90.21
	Final Model (Test)	94.35	94.35	94.35	92.81

The performance results for Dataset 1 and Dataset 2, respectively, are summarized in Table 7. The model reaches an accuracy of 96.87% on the validation set and 96.52% on the test set, along with high maF1, MCC, and Kappa values for Dataset 1. The close agreement between validation and test results indicates stable learning behavior and minimal performance degradation when applied to unseen samples. A similar trend is observed for Dataset 2, the Final Model achieves an accuracy of 93.40% on the validation set and 94.35% on the test set, demonstrating that the learned representations generalize effectively to different data distributions. The agreement metrics further indicate consistent class predictions and stable decision boundaries.

4.5 Complexity Analysis of Ensemble Guided Pseudo Labeling Models

In addition to predictive performance, the computational efficiency of the proposed framework is assessed using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), training time, and inference time.

Table 8 summarizes the efficiency analysis for both Dataset 1 and Dataset 2 across the Initial, Retrained, and Final Models.

Table 8. Computational efficiency analysis across Dataset 1 and Dataset 2.

Dataset	Model	RMSE	MAE	Training Time (s)	Inference Time (s)
Dataset 1	Initial Model	0.3974 ± 0.0632	0.1611 ± 0.0534	3.0371 ± 0.4470	0.0126 ± 0.0087
	Retrained Model	0.2461 ± 0.0629	0.0637 ± 0.0322	15.0081 ± 0.9248	0.0246 ± 0.0192
	Final Model	0.1664 ± 0.0246	0.0282 ± 0.0081	25.2492 ± 0.6645	0.0463 ± 0.0226
Dataset 2	Initial Model	0.4364 ± 0.0645	0.1938 ± 0.0559	0.9205 ± 0.0940	0.0269 ± 0.0145
	Retrained Model	0.3613 ± 0.0382	0.1317 ± 0.0278	0.8687 ± 0.1721	0.0240 ± 0.0180
	Final Model	0.2907 ± 0.1001	0.0925 ± 0.0429	1.1116 ± 0.5903	0.0153 ± 0.0136

For Dataset 1, error metrics decrease progressively from the Initial Model to the Final Model, with the Final Model achieving the lowest RMSE (0.1664 ± 0.0246) and MAE (0.0282 ± 0.0081). Although the Final Model requires a longer training time (25.25 ± 0.66 s), its inference time remains low (0.0463 ± 0.23 s). Dataset 2, exhibits a similar pattern, where the Final Model attains the lowest RMSE (0.2907 ± 0.1001) and MAE (0.0925 ± 0.0429), highlighting the effectiveness of iterative pseudo-labeling and retraining. While additional SSL optimization increases training duration, inference remains efficient, resulting in minimal deployment cost. These results demonstrate a favorable accuracy–efficiency tradeoff, with substantial error reduction achieved at a modest increase in computational effort.

4.6 Feature-group ablation study

To examine the contribution of different feature groups to the proposed framework, a comprehensive ablation study is conducted separately on Dataset 1 and Dataset 2. In this study, selected feature groups are systematically excluded while retaining the remaining features, enabling analysis of their individual and combined impact on classification performance. Evaluation is performed using multiple metrics, including Accuracy, maP, maR, maF1, MCC, Kappa, RMSE, and MAE.

The ablation results presented in Table 9 reveal clear and interpretable trends regarding the contribution of individual feature groups. Removing the TQWT features leads to a substantial performance decline, with MCC dropping from 89.94% to 77.13%, accuracy from 97.18% to 95.71%, and maF1 from 94.69% to 87.71%, demonstrating that TQWT captures highly discriminative nonlinear time-frequency features essential for PD classification. Excluding MFCC features also results in notable performance degradation, particularly in MCC and recall-related metrics, highlighting that MFCC provides complementary spectral information that enhances class separability when combined with nonlinear representations. Conversely, removing wavelet features slightly improves performance (MCC 89.94% to 91.02%; maF1 94.69% to 95.35%) and reduces training time, suggesting potential redundancy or noise introduced by these features. Exclusion of baseline or other time-frequency features produces moderate performance decay, indicating a supportive but non-dominant role in stabilizing classification.

For Dataset 2, the exclusion of Temporal or MFCC feature groups results in consistent performance degradation across key evaluation metrics for the Final Model. Specifically, removing MFCC features reduces accuracy from 92.24% to 88.62%, MCC from 73.12% to 61.22%, and maF1 from 85.00% to 78.56%. In contrast, eliminating Noise or Nonlinear feature groups improves performance across the feature set. For example, omitting Noise features increases accuracy from 92.24% to 92.67% and MCC from 73.12% to 74.69%, while removing Nonlinear features further enhances performance, achieving an MCC of 80.19% and maF1 of 89.68%. These observations suggest that certain feature groups may introduce redundancy or dataset-specific noise that negatively impacts generalization.

Overall, across both datasets, while the full feature set provides competitive performance, carefully selected reduced feature combinations consistently outperform it, suggesting that strategic feature fusion is more effective than exhaustive feature aggregation. The consistent ablation patterns observed in the Initial, Retrained, Final, and Global Models indicate that the proposed framework preserves stable and interpretable feature importance relationships.

Table 9. Feature-group ablation results on Dataset 1 across multiple performance metrics, with the top three results highlighted in green, blue, and red.

Excluded Feature	Model	Accuracy (%)	maP (%)	maR (%)	maF1 (%)	MCC (%)	Kappa (%)	RMSE	MAE
TQWT	Initial Model	81.67 $\pm$ 5.41	73.61 $\pm$ 10.87	65.89 $\pm$ 10.29	67.39 $\pm$ 11.71	38.36 $\pm$ 21.34	36.35 $\pm$ 21.91	0.4242 $\pm$ 0.0647	0.1833 $\pm$ 0.0541
	Retrained Model	92.50 $\pm$ 2.89	87.05 $\pm$ 8.37	75.71 $\pm$ 7.47	79.75 $\pm$ 7.84	61.54 $\pm$ 15.27	59.75 $\pm$ 15.55	0.2689 $\pm$ 0.0581	0.0750 $\pm$ 0.0289
	Final Model	95.71 $\pm$ 1.01	95.52 $\pm$ 3.46	82.70 $\pm$ 2.89	87.71 $\pm$ 2.98	77.13 $\pm$ 5.89	75.51 $\pm$ 5.92	0.2059 $\pm$ 0.0242	0.0429 $\pm$ 0.0101
	Global Model	95.69	94.49	95.69	95.69	95.69	94.49	0.0431	0.0431
Baseline	Initial Model	83.89 $\pm$ 6.02	78.85 $\pm$ 14.54	68.21 $\pm$ 11.13	70.59 $\pm$ 13.07	45.60 $\pm$ 25.10	42.73 $\pm$ 24.72	0.3961 $\pm$ 0.0729	0.1611 $\pm$ 0.0602
	Retrained Model	95.00 $\pm$ 2.15	95.90 $\pm$ 2.68	86.10 $\pm$ 5.81	89.91 $\pm$ 4.81	81.29 $\pm$ 8.56	79.94 $\pm$ 9.50	0.2195 $\pm$ 0.0478	0.0500 $\pm$ 0.0215
	Final Model	97.34 $\pm$ 0.38	97.93 $\pm$ 0.79	92.39 $\pm$ 1.19	94.88 $\pm$ 0.79	90.14 $\pm$ 1.46	89.77 $\pm$ 1.57	0.1628 $\pm$ 0.0117	0.0266 $\pm$ 0.0038
	Global Model	96.97	96.43	96.97	96.97	96.97	96.43	0.0303	0.0303
Time Frequency	Initial Model	84.44 $\pm$ 5.41	78.67 $\pm$ 12.71	70.36 $\pm$ 10.25	72.72 $\pm$ 12.00	48.25 $\pm$ 22.87	46.49 $\pm$ 22.65	0.3903 $\pm$ 0.0637	0.1556 $\pm$ 0.0541
	Retrained Model	94.55 $\pm$ 1.93	94.84 $\pm$ 2.91	84.67 $\pm$ 4.88	88.68 $\pm$ 4.29	78.80 $\pm$ 7.94	77.48 $\pm$ 8.49	0.2303 $\pm$ 0.0437	0.0545 $\pm$ 0.0193
	Final Model	97.52 $\pm$ 0.52	97.72 $\pm$ 0.84	92.57 $\pm$ 1.91	94.89 $\pm$ 1.16	90.12 $\pm$ 2.15	89.80 $\pm$ 2.31	0.1567 $\pm$ 0.0166	0.0248 $\pm$ 0.0052
	Global Model	96.73	96.13	96.73	96.73	96.73	96.13	0.0327	0.0327
Vocal Fold	Initial Model	83.89 $\pm$ 5.34	79.32 $\pm$ 14.01	68.21 $\pm$ 9.21	70.96 $\pm$ 11.35	46.14 $\pm$ 22.58	43.26 $\pm$ 21.51	0.3974 $\pm$ 0.0632	0.1611 $\pm$ 0.0534
	Retrained Model	93.58 $\pm$ 2.59	93.70 $\pm$ 4.11	83.24 $\pm$ 6.27	87.18 $\pm$ 5.67	76.11 $\pm$ 10.40	74.53 $\pm$ 11.19	0.2492 $\pm$ 0.0513	0.0642 $\pm$ 0.0259
	Final Model	97.04 $\pm$ 0.70	97.29 $\pm$ 1.24	92.41 $\pm$ 1.76	94.61 $\pm$ 1.32	89.55 $\pm$ 2.53	89.23 $\pm$ 2.64	0.1710 $\pm$ 0.0216	0.0296 $\pm$ 0.0070
	Global Model	96.96	95.39	96.96	96.96	96.96	95.39	0.0304	0.0304
MFCC	Initial Model	83.89 $\pm$ 7.19	77.58 $\pm$ 14.99	69.11 $\pm$ 12.70	71.30 $\pm$ 14.35	45.78 $\pm$ 27.60	43.84 $\pm$ 27.51	0.3915 $\pm$ 0.0989	0.1611 $\pm$ 0.0719
	Retrained Model	94.80 $\pm$ 2.59	95.19 $\pm$ 2.42	86.11 $\pm$ 7.85	89.42 $\pm$ 6.05	80.47 $\pm$ 10.29	78.99 $\pm$ 11.91	0.2224 $\pm$ 0.0562	0.0520 $\pm$ 0.0259
	Final Model	97.39 $\pm$ 1.00	97.58 $\pm$ 1.90	92.33 $\pm$ 2.24	94.71 $\pm$ 2.05	89.75 $\pm$ 4.04	89.43 $\pm$ 4.09	0.1591 $\pm$ 0.0314	0.0261 $\pm$ 0.0100
	Global Model	95.44	94.64	95.44	95.44	95.44	94.64	0.0456	0.0456
Wavelet	Initial Model	87.22 $\pm$ 5.76	83.56 $\pm$ 11.58	75.71 $\pm$ 12.81	77.45 $\pm$ 13.94	58.11 $\pm$ 24.46	56.11 $\pm$ 25.74	0.3510 $\pm$ 0.0759	0.1278 $\pm$ 0.0576
	Retrained Model	97.01 $\pm$ 0.78	98.25 $\pm$ 0.44	91.58 $\pm$ 2.20	94.49 $\pm$ 1.53	89.56 $\pm$ 2.78	88.99 $\pm$ 3.05	0.1717 $\pm$ 0.0234	0.0299 $\pm$ 0.0078
	Final Model	97.48 $\pm$ 0.85	97.82 $\pm$ 1.12	93.34 $\pm$ 2.66	95.35 $\pm$ 1.69	91.02 $\pm$ 3.08	90.70 $\pm$ 3.37	0.1570 $\pm$ 0.0272	0.0252 $\pm$ 0.0085
	Global Model	97.31	96.58	97.31	97.31	97.31	96.58	0.0269	0.0269
All	Initial Model	83.89 $\pm$ 5.34	79.32 $\pm$ 14.01	68.21 $\pm$ 9.21	70.96 $\pm$ 11.35	46.14 $\pm$ 22.58	43.26 $\pm$ 21.51	0.3974 $\pm$ 0.0632	0.1611 $\pm$ 0.0534
	Retrained Model	93.63 $\pm$ 3.22	92.68 $\pm$ 6.81	83.57 $\pm$ 6.35	87.22 $\pm$ 6.50	75.67 $\pm$ 12.92	74.54 $\pm$ 12.94	0.2461 $\pm$ 0.0629	0.0637 $\pm$ 0.0322
	Final Model	97.18 $\pm$ 0.81	98.36 $\pm$ 0.46	91.84 $\pm$ 2.35	94.69 $\pm$ 1.63	89.94 $\pm$ 2.96	89.41 $\pm$ 3.25	0.1664 $\pm$ 0.0246	0.0282 $\pm$ 0.0081
	Global Model	97.14	96.43	97.14	97.14	97.14	96.43	0.0286	0.0286

Table 10. Feature-group ablation results on Dataset 2 across multiple performance metrics, with the top three results highlighted in green, blue, and red.

Excluded Feature	Model	Accuracy (%)	maP (%)	maR (%)	maF1 (%)	MCC (%)	Kappa (%)	RMSE	MAE
Acoustic	Initial Model	82.50 $\pm$ 3.56	84.57 $\pm$ 5.75	68.33 $\pm$ 9.01	69.87 $\pm$ 10.51	48.29 $\pm$ 12.48	42.49 $\pm$ 17.71	0.4165 $\pm$ 0.0435	0.1750 $\pm$ 0.0356
	Retrained Model	88.85 $\pm$ 1.61	86.89 $\pm$ 4.63	71.29 $\pm$ 5.28	75.45 $\pm$ 5.10	55.64 $\pm$ 7.89	51.74 $\pm$ 9.61	0.3333 $\pm$ 0.0239	0.1115 $\pm$ 0.0161
	Final Model	92.26 $\pm$ 2.33	93.26 $\pm$ 4.14	78.99 $\pm$ 6.44	83.61 $\pm$ 5.75	70.54 $\pm$ 9.84	67.61 $\pm$ 11.21	0.2755 $\pm$ 0.0439	0.0774 $\pm$ 0.0233
	Global Model	95.86	94.72	95.86	95.86	95.86	94.72	0.0414	0.0414
Noise	Initial Model	83.12 $\pm$ 3.56	83.67 $\pm$ 3.83	69.58 $\pm$ 8.67	71.49 $\pm$ 10.35	49.76 $\pm$ 12.82	45.21 $\pm$ 17.53	0.4090 $\pm$ 0.0432	0.1688 $\pm$ 0.0356
	Retrained Model	85.77 $\pm$ 3.99	81.84 $\pm$ 10.31	69.86 $\pm$ 4.90	73.03 $\pm$ 5.54	49.87 $\pm$ 12.46	46.76 $\pm$ 11.23	0.3746 $\pm$ 0.0496	0.1423 $\pm$ 0.0399
	Final Model	92.67 $\pm$ 1.27	92.65 $\pm$ 1.79	82.84 $\pm$ 4.52	86.43 $\pm$ 2.98	74.69 $\pm$ 4.84	73.05 $\pm$ 5.84	0.2700 $\pm$ 0.0238	0.0733 $\pm$ 0.0127
	Global Model	96.08	94.00	96.08	96.08	96.08	94.00	0.0392	0.0392
Non Linear	Initial Model	79.38 $\pm$ 7.19	73.28 $\pm$ 16.19	65.42 $\pm$ 11.08	66.80 $\pm$ 12.81	37.55 $\pm$ 25.69	35.20 $\pm$ 24.47	0.4485 $\pm$ 0.0798	0.2062 $\pm$ 0.0719
	Retrained Model	86.12 $\pm$ 3.93	82.43 $\pm$ 14.39	66.53 $\pm$ 10.43	69.08 $\pm$ 11.31	45.09 $\pm$ 21.06	40.02 $\pm$ 21.28	0.3698 $\pm$ 0.0503	0.1388 $\pm$ 0.0393
	Final Model	93.13 $\pm$ 1.11	93.13 $\pm$ 1.84	87.34 $\pm$ 2.92	89.68 $\pm$ 1.86	80.19 $\pm$ 3.35	79.42 $\pm$ 3.69	0.2613 $\pm$ 0.0220	0.0687 $\pm$ 0.0111
	Global Model	95.91	94.00	95.91	95.91	95.91	94.00	0.0409	0.0409
MFCC	Initial Model	81.88 $\pm$ 5.13	84.81 $\pm$ 10.39	67.92 $\pm$ 8.15	69.65 $\pm$ 9.92	48.11 $\pm$ 14.99	41.87 $\pm$ 17.28	0.4224 $\pm$ 0.0563	0.1571 $\pm$ 0.0433
	Retrained Model	84.29 $\pm$ 4.33	76.04 $\pm$ 6.71	63.53 $\pm$ 14.43	64.96 $\pm$ 14.43	36.32 $\pm$ 9.55	32.43 $\pm$ 9.38	0.3932 $\pm$ 0.0563	0.1574 $\pm$ 0.0433
	Final Model	88.62 $\pm$ 2.32	88.66 $\pm$ 6.45	74.42 $\pm$ 4.33	78.56 $\pm$ 4.59	61.22 $\pm$ 8.86	57.75 $\pm$ 8.90	0.3360 $\pm$ 0.00344	0.1138 $\pm$ 0.0233
	Global Model	95.27	91.61	95.27	95.27	95.27	91.61	0.0473	0.0473
Temporal	Initial Model	83.12 $\pm$ 4.19	82.19 $\pm$ 7.69	71.25 $\pm$ 7.57	73.33 $\pm$ 7.92	51.41 $\pm$ 12.31	47.95 $\pm$ 14.55	0.4082 $\pm$ 0.0518	0.1688 $\pm$ 0.0419
	Retrained Model	87.45 $\pm$ 2.97	81.72 $\pm$ 9.19	71.43 $\pm$ 7.05	74.39 $\pm$ 6.91	51.74 $\pm$ 13.87	49.36 $\pm$ 13.55	0.3523 $\pm$ 0.0418	0.1255 $\pm$ 0.0297
	Final Model	92.58 $\pm$ 1.34	91.56 $\pm$ 2.25	88.73 $\pm$ 1.74	89.98 $\pm$ 1.78	80.23 $\pm$ 3.58	79.98 $\pm$ 3.55	0.2714 $\pm$ 0.0261	0.0742 $\pm$ 0.0134
	Global Model	94.94	94.00	94.94	94.94	94.94	94.00	0.0506	0.0506
All	Initial Model	80.62 $\pm$ 5.59	73.77 $\pm$ 9.81	67.92 $\pm$ 11.18	68.82 $\pm$ 12.54	40.88 $\pm$ 21.71	39.34 $\pm$ 22.88	0.4364 $\pm$ 0.0645	0.1938 $\pm$ 0.0559
	Retrained Model	86.83 $\pm$ 2.78	83.66 $\pm$ 6.91	70.98 $\pm$ 7.87	74.02 $\pm$ 7.66	52.53 $\pm$ 12.99	49.07 $\pm$ 14.44	0.3613 $\pm$ 0.0382	0.1347 $\pm$ 0.0371
	Final Model	92.24 $\pm$ 4.29	92.72 $\pm$ 5.04	81.48 $\pm$ 9.33	85.00 $\pm$ 8.20	73.12 $\pm$ 14.70	70.46 $\pm$ 16.08	0.290700 $\pm$ 0.01001	0.0925 $\pm$ 0.0429
	Global Model	95.50	93.00	95.50	95.50	95.50	93.05	0.0452	0.0388

Fig 5. MCC variation with feature group removal in Dataset 1.

Fig 6. MCC variation with feature group removal in Dataset 2.

The change in MCC (ΔMCC) obtained by removing individual feature groups from the full feature set, as defined in Eq. 24.

$$\Delta\text{MCC} = \text{MCC}_{\text{except}} - \text{MCC}_{\text{full}} \quad (24)$$

Using the full dataset, the model achieves an MCC of 97.14%, serving as the reference baseline. The largest performance decline occurs when the MFCC feature group is excluded, causing MCC to drop from 97.14% to 95.44% ($\Delta\text{MCC} = -1.70\%$), representing the most substantial negative impact among all feature groups. Similarly, removing TQWT features reduces MCC from 97.14% to 95.69% ($\Delta\text{MCC} = -1.45\%$), highlighting TQWT’s role in capturing highly discriminative time-frequency characteristics related to PD vocal impairments. In contrast, excluding the Wavelet feature group slightly improves MCC ($\Delta\text{MCC} = +0.17\%$), suggesting potential redundancy or noise when combined with more expressive representations such as TQWT and MFCC. Fig. 5 clearly illustrates that TQWT and MFCC are the most critical feature sets in the proposed framework, as the absence of either results in a marked decline in classification performance.

As illustrated in Fig. 6, the greatest performance decline for Dataset 2 occurs when Temporal features are excluded, with MCC dropping from 95.50% to 94.94%, highlighting the importance of temporal dynamics in capturing disease-related variations. Similarly, excluding MFCC features reduces MCC from 95.50% to 95.27%, confirming their critical role in effective PD detection. In contrast, removal of Acoustic, Noise, or Nonlinear feature groups results in minor performance improvements. These findings indicate that MFCC and Temporal features are the most discriminative components for Dataset 2.

Overall, the ablation analysis highlights that MFCC features are consistently critical across both datasets, while TQWT features play a key role in Dataset 1 and Temporal features are most important in Dataset 2. Feature groups whose removal results in a positive ΔMCC are comparatively less informative, whereas those producing a negative ΔMCC are essential for achieving robust and reliable classification performance.

4.7 Comparison with Existing SSL models

Following the experimental evaluation of the proposed framework, its effectiveness was further assessed by benchmarking against several state-of-the-art SSL models on the same datasets [59]. The comparison included Rasco [60] and RelRasco [61], which employ ensemble strategies combined with subspace selection, as well as CoTrainingByCommittee [62] and CoTraining [63], both of which expand the labeled set using multiple classifiers or views. Additionally, Setred [64] incorporates a rejection mechanism to filter noisy predictions, while DeTriTraining [65] leverages consensus among multiple classifiers to assign labels to unlabeled data. Other approaches, including CoForest [66], TriTraining [67], DemocraticCoLearning [68], and SelfTraining [69], utilize ensemble agreement rules, diverse classifier integration, or iterative high-confidence retraining to enhance labeling reliability.

Table 11. Comparison of the proposed SSL model awith baseline SSL models on Dataset 1 (20% labeled data).

Model	Accuracy (%)	maF1 (%)	ROC-AUC (%)	MCC (%)	Training Time (s)
Rasco [60]	0.8440 ± 0.0349	0.8438 ± 0.0350	0.9179 ± 0.0281	0.6887 ± 0.0690	16.2906 ± 0.6731
RelRasco [61]	0.8413 ± 0.0483	0.8411 ± 0.0484	0.9152 ± 0.0360	0.6843 ± 0.0963	17.3634 ± 1.1337
CoTrainingByCommittee [62]	0.7704 ± 0.0238	0.7693 ± 0.0237	0.8493 ± 0.0449	0.5461 ± 0.0487	27.1915 ± 2.6403
CoTraining [63]	0.7429 ± 0.0189	0.7422 ± 0.0189	0.7429 ± 0.0189	0.4888 ± 0.0382	6.6953 ± 0.3016
Setred [64]	0.7420 ± 0.0322	0.7411 ± 0.0326	0.7419 ± 0.0323	0.4872 ± 0.0639	1.9757 ± 0.3437
DeTriTraining [65]	0.6924 ± 0.0239	0.6921 ± 0.0241	0.6924 ± 0.0240	0.3856 ± 0.0477	220.0038 ± 31.8402
CoForest [66]	0.6862 ± 0.0369	0.6853 ± 0.0372	0.7071 ± 0.0347	0.3745 ± 0.0735	9.9148 ± 1.3377
TriTraining [67]	0.7660 ± 0.0201	0.7635 ± 0.0214	0.8643 ± 0.0221	0.5426 ± 0.0388	0.2983 ± 0.0367
Democratic Co-Learning [68]	0.7544 ± 0.0469	0.7500 ± 0.0499	0.8214 ± 0.0241	0.5238 ± 0.0846	8.7140 ± 1.3751
Self-Training [69]	0.5000 ± 0.0000	0.5000 ± 0.0000	0.5000 ± 0.0000	0.6426 ± 0.0388	8.9148 ± 1.3378
Proposed SSL	0.9673 ± 0.0115	0.9415 ± 0.0206	0.7816 ± 0.0315	0.8871 ± 0.0410	28.2966 ± 0.5478

Table 11 shows the evaluation of the proposed SSL model against SSL baselines on Dataset 1 with 20% labeled data. TriTraining demonstrates the lowest training time (0.2983 ± 0.0367 s) due to its simple agreement-based pseudo-labeling without iterative ensemble refinement; however, this efficiency comes at the cost of noticeably lower accuracy, maF1, MCC, and ROC-AUC compared to the proposed SSL model. In contrast, DeTriTraining incurs the highest training time (220.0038 ± 31.8402 s) owing to repeated disagreement resolution and iterative retraining across multiple classifiers, yet fails to achieve competitive performance across all evaluation metrics. Among the baseline SSL approaches, Rasco and RelRasco achieve the closest accuracy to the proposed model, with 84.4% (± 0.0349) and 84.13% (± 0.0483), respectively. Nevertheless, the proposed SSL framework delivers a substantial relative improvement of approximately 12.3% over Rasco and 12.6% over RelRasco. While Rasco reports higher ROC-AUC values exceeding 91%, this advantage does not translate into superior class-wise predictive performance. The proposed SSL incurs a moderately higher training time (28.2966 ± 0.5478 s) due to ensemble-guided consensus formation, yet it remains significantly more efficient than DeTriTraining and achieves the best overall performance.

Table 12. Comparison of the proposed SSL model with baseline SSL models on Dataset 1 (40% labeled data).

Model	Accuracy (%)	maF1 (%)	ROC-AUC (%)	MCC (%)	Training Time (s)
Rasco [60]	0.8750 ± 0.0372	0.8748 ± 0.0375	0.9504 ± 0.0219	0.7518 ± 0.0732	40.8378 ± 2.0202
RelRasco [61]	0.8794 ± 0.0304	0.8793 ± 0.0305	0.9507 ± 0.0242	0.7606 ± 0.0599	42.6047 ± 1.0982
CoTrainingByCommittee [62]	0.8378 ± 0.0202	0.8370 ± 0.0202	0.9169 ± 0.0197	0.6821 ± 0.0414	23.0860 ± 2.5534
CoTraining [63]	0.8156 ± 0.0439	0.8147 ± 0.0448	0.8155 ± 0.0441	0.6358 ± 0.0849	16.1456 ± 1.0803
Setred [64]	0.8103 ± 0.0474	0.8099 ± 0.0474	0.8103 ± 0.0474	0.6233 ± 0.0948	2.0380 ± 0.3444
DeTriTraining [65]	0.6968 ± 0.0267	0.6963 ± 0.0269	0.6969 ± 0.0267	0.3949 ± 0.0532	241.5351 ± 43.9557
CoForest [66]	0.7863 ± 0.0123	0.7859 ± 0.0126	0.7958 ± 0.0205	0.5749 ± 0.0234	16.6340 ± 0.6212
TriTraining [67]	0.8156 ± 0.0301	0.8139 ± 0.0309	0.9012 ± 0.0181	0.6425 ± 0.0590	0.5958 ± 0.0736
Democratic Co-Learning [68]	0.8005 ± 0.0308	0.7988 ± 0.0314	0.8541 ± 0.0345	0.6110 ± 0.0618	7.4311 ± 0.8258
Self-Training [69]	0.5000 ± 0.0000	0.5000 ± 0.0000	0.5000 ± 0.0000	0.6541 ± 0.0427	0.6540 ± 0.0045
Proposed SSL	0.9674 ± 0.0086	0.9674 ± 0.0086	0.9541 ± 0.0345	0.9350 ± 0.0171	25.90 ± 0.3193

A similar trend is observed on Dataset 1 with a 40% labeled set, as summarized in Table 12. The proposed SSL model consistently outperforms all baseline methods, achieving the highest accuracy of 96.74% (± 0.0086), maF1 of 96.74% (± 0.0086), and MCC of 93.50% (± 0.0345), while maintaining a moderate training time of 25.90 ± 0.3193 s. TriTraining again exhibits the lowest training time (0.5958 ± 0.0736 s) due to its simple agreement-based pseudo-labeling; however, this comes with reduced accuracy, MCC, and ROC-AUC. In contrast, DeTriTraining is the most computationally expensive method (241.5351 ± 43.9557 s) yet delivers the poorest performance, highlighting that longer training does not necessarily improve results. Among the baseline SSL approaches, Rasco and RelRasco achieve the closest performance to the proposed model, with accuracies of 87.50% (± 0.0086) and 87.94% (± 0.0304), respectively, along with comparable maF1 scores. Nevertheless, the proposed SSL model maintains the best overall performance, providing a relative accuracy improvement of nearly 9% and superior balance across evaluation metrics.

Table 13. Comparison of the proposed SSL model with baseline SSL models on Dataset 2 (20% labeled data).

Model	Accuracy (%)	maF1 (%)	ROC-AUC (%)	MCC (%)	Training Time (s)
Rasco [60]	0.8386 ± 0.0107	0.8384 ± 0.0106	0.9131 ± 0.0153	0.6792 ± 0.0228	0.9868 ± 0.1776
RelRasco [61]	0.8371 ± 0.0153	0.8370 ± 0.0151	0.9149 ± 0.0160	0.6755 ± 0.0317	1.1205 ± 0.2364
CoTrainingByCommittee [62]	0.8100 ± 0.0357	0.8096 ± 0.0359	0.8613 ± 0.0298	0.6224 ± 0.0708	0.6750 ± 0.0979
CoTraining [63]	0.7614 ± 0.0340	0.7612 ± 0.0340	0.7614 ± 0.0340	0.5235 ± 0.0682	0.3664 ± 0.0379
Setred [64]	0.7800 ± 0.0464	0.7797 ± 0.0467	0.7798 ± 0.0466	0.5608 ± 0.0926	0.2015 ± 0.0102
DeTriTraining [65]	0.8300 ± 0.0249	0.8298 ± 0.0249	0.8300 ± 0.0249	0.6618 ± 0.0503	7.0991 ± 0.9318
CoForest [66]	0.7714 ± 0.0386	0.7712 ± 0.0386	0.7716 ± 0.0386	0.5443 ± 0.0775	1.1337 ± 0.3333
TriTraining [67]	0.8304 ± 0.0425	0.8294 ± 0.0431	0.9160 ± 0.0282	0.6633 ± 0.0824	0.0269 ± 0.0028
Democratic Co-Learning [68]	0.8386 ± 0.0471	0.8382 ± 0.0475	0.8539 ± 0.0400	0.6794 ± 0.0923	1.6873 ± 0.5474
Self-Training [69]	0.5000 ± 0.0000	0.5000 ± 0.0000	0.5000 ± 0.0000	0.6454 ± 0.0824	0.6873 ± 0.5474
Proposed SSL	0.9639 ± 0.0045	0.9640 ± 0.0023	0.9096 ± 0.0368	0.1180 ± 0.1712	1.1353 ± 0.0368

On Dataset 2 with 20% labeled data, the proposed SSL model is evaluated against several state-of-the-art SSL baselines, as summarized in Table 13. The SSL model achieves the highest accuracy of 96.39%

(± 0.0045), representing a substantial improvement over all existing methods, including Rasco and RelRasco, which are 12–14% less accurate. While TriTraining offers the lowest training time (0.0269 ± 0.0028 s) due to its simple agreement-based pseudo-labeling, it does not provide competitive predictive performance. Conversely, DeTriTraining incurs the highest training time (7.0991 ± 0.9318 s) without yielding better results. Although Rasco and RelRasco attain relatively high maF1 and ROC-AUC scores, their overall accuracy remains notably below that of the proposed SSL model.

Table 14. Comparison of the proposed SSL model with baseline SSL models on Dataset 2 (40% labeled data).

Model	Accuracy (%)	maF1 (%)	ROC-AUC (%)	MCC (%)	Training Time (s)
Rasco [60]	0.8929 ± 0.0256	0.8928 ± 0.0256	0.9521 ± 0.0130	0.7863 ± 0.0510	1.8171 ± 0.2114
RelRasco [61]	0.8843 ± 0.0365	0.8842 ± 0.0366	0.9469 ± 0.0186	0.7693 ± 0.0725	1.9883 ± 0.3283
CoTrainingByCommittee [62]	0.8543 ± 0.0366	0.8542 ± 0.0366	0.9051 ± 0.0246	0.7097 ± 0.0734	0.7479 ± 0.1577
CoTraining [63]	0.8157 ± 0.0165	0.8155 ± 0.0166	0.8158 ± 0.0165	0.6330 ± 0.0328	0.6725 ± 0.0496
Setred [64]	0.8314 ± 0.0390	0.8313 ± 0.0392	0.8316 ± 0.0387	0.6642 ± 0.0764	0.1956 ± 0.0121
DeTriTraining [65]	0.8200 ± 0.0311	0.8198 ± 0.0310	0.8200 ± 0.0311	0.6417 ± 0.0629	8.6000 ± 0.7904
CoForest [66]	0.8086 ± 0.0457	0.8082 ± 0.0460	0.8163 ± 0.0267	0.6195 ± 0.0896	3.4631 ± 0.3153
TriTraining [67]	0.8714 ± 0.0217	0.8713 ± 0.0218	0.9446 ± 0.0060	0.7441 ± 0.0427	0.0520 ± 0.0085
Democratic Co-Learning [68]	0.8586 ± 0.0261	0.8585 ± 0.0261	0.8926 ± 0.0326	0.7180 ± 0.0521	1.1115 ± 0.2390
Self-Training [69]	0.5000 ± 0.0000	0.5000 ± 0.0000	0.5000 ± 0.0000	0.6541 ± 0.0427	0.0540 ± 0.0045
Proposed SSL	0.8990 ± 0.0256	0.8987 ± 0.0257	0.8551 ± 0.0324	0.7988 ± 0.0509	1.4698 ± 0.3842

Finally, the evaluation on Dataset 2 with 40% labeled data, summarized in Table 14, further demonstrates the effectiveness of the proposed framework. The SSL model achieves the highest accuracy and maF1, reaching 89.90% (± 0.0256) and 89.87% (± 0.0257), respectively, outperforming all baseline models, including Rasco and RelRasco. It also attains the best MCC of 79.88% (± 0.0509), indicating strong class-wise predictive capability. The ROC-AUC of 85.51% (± 0.0324) is slightly lower than Rasco’s 95.21% (± 0.0130), reflecting that the model emphasizes a voting-based ensemble approach rather than fully probabilistic rankings. TriTraining again exhibits the fastest training time (0.0520 ± 0.0085 s), whereas DeTriTraining is the most computationally expensive (8.6000 ± 0.7904 s) without achieving notable gains in accuracy or MCC. Rasco and RelRasco provide high ROC-AUC and moderate accuracy but remain inferior to the proposed SSL model in terms of balanced predictive performance.

4.8 SHAP: global interpretability

To improve the interpretability of the proposed framework, SHAP is used on the global model to measure each feature’s contribution to predictions. In this study, SHAP analysis is conducted on representative samples from the test set, enabling both instance-level and global-level insights into model behavior. By evaluating the marginal contribution of each feature, SHAP reveals how individual features influence predictions across the ensemble, offering a transparent understanding of the decision-making process.

Fig 7. SHAP waterfall plots for Dataset 1: (a) XGB, (b) GB, and (c) LGBM.

Fig. 7 illustrates SHAP waterfall plots for Dataset 1 using three ensemble-guided models: (a) XGB, (b) GB, and (c) LGBM. Each plot depicts the contribution of individual features to the model’s predictions, where red bars indicate positive contributions that increase the predicted outcome, and blue bars indicate negative contributions that decrease it. In Fig. 7(a), the XGB model highlights features such as `app_det_TKE0_mean_1_coef`, `std_MFCC_6th_coef`, and `mean_MFCC_3nd_coef` as having the most significant influence. In the GB model, `tqwt_entropy_log_dec_35` contributes positively, as shown in Fig. 7(b). In Fig. 7(c), the LGBM model emphasizes features such as `tqwt_maxValue_dec_14`, and `tqwt_kurtosisValue_dec_17` as key contributors to model output. These plots provide insights into how each model interprets the underlying data, explaining and visualizing the decision-making process for PD classification. Overall, the SHAP waterfall plots indicate that MFCC and TQWT features consistently contribute positively across models, highlighting their critical role in enhancing both model interpretability and reliability.

Fig 8. SHAP waterfall plots for Dataset 2: (a) XGB, (b) GB, and (c) LGBM.

Table 15. Comparison of the proposed FEP-SSL with existing models.

Reference	Method	Accuracy (%)	Precision (%)	MCC (%)	F1-score (%)
Bao et al. [23]	SSL	83.8%	-	67.5%	-
Danek et al. [27]	FL	87.6%	96.02%	0.502	-
Adnan et al. [70]	SSL	85.7%	81.08%	97%	77.93%
Egbo et al. [71]	XGB with Borderline SMOTE	92%	64%	100%	95%
Proposed FEP-SSL (Dataset 1)	SSL + FL	97.14%	96.43%	97.14%	97.14%

For Dataset 2, the SHAP waterfall plots of three ensemble-guided models—XGB, GB, and LGBM—are presented in Fig. 8. In Fig. 8(a), the XGB model’s predictions are primarily influenced by voice-related features such as **HNR38**, **Gender**, and **Shim_dB**, which contribute negatively and thus reduce the predicted score. In contrast, **RPDE**, **HNR35**, and **Jitter_abs** contribute positively, partially offsetting the negative effects. Fig. 8(b) illustrates the GB model, where features including **MFCC9**, **MFCC10**, and **GNE** have the strongest positive contributions. Several MFCC-related features jointly impact the output, emphasizing the importance of spectral information for prediction. In Fig. 8(c), the LGBM model highlights Delta-based features, with **Delta12**, **Delta7**, and **Delta10** contributing positively, whereas **Delta1** and **Delta9** have negative effects. This indicates that temporal variations in speech play a critical role in the decision-making process of LGBM. Overall, the SHAP analysis for Dataset 2 demonstrates that MFCC and Temporal features are the most influential across all models. These features consistently dominate predictions, suggesting that spectral and temporal speech dynamics are more informative than static voice characteristics for reliable model decision-making.

4.9 Comparison with existing literature

Recent research has examined combining SSL with FL frameworks for PD classification. Bao et al. [23] employed an SSL framework, achieving 83.8% accuracy and 67.5% MCC, although some evaluation metrics were not reported. The reliance on handcrafted features and linear dimensionality reduction constrained class separability, as reflected in the moderate MCC. Danek et al. [27] explored a federated learning approach, reporting a ROC-AUC of 87.6% and precision of 96.02%, demonstrating the privacy-preserving benefits of distributed training. However, high precision led to reduced recall, and FL introduced slight performance degradation. Adnan et al. [70] reported an SSL model achieving 85.7% accuracy and an F1-score of 77.93%, highlighting the benefits of leveraging unlabeled data. However, performance was limited by class imbalance and environmental variability, resulting in reduced sensitivity and lower maF1 scores. Egbo et al. [71] applied an XGB model with Borderline-SMOTE, achieving 92% accuracy and an F1-score of 95%. In comparison, the proposed framework, which integrates SSL with FL, outperforms all prior methods across key metrics. It achieves 97.14% accuracy, 96.43% precision, and 97.12% MCC, demonstrating superior class balance, stronger generalization, and robust performance in privacy-sensitive healthcare settings. These results substantially exceed the performance of existing approaches, as summarized in Table 15.

4.10 Discussion

The proposed FEP-SSL framework demonstrates superior performance, particularly in scenarios characterized by limited labeled data and stringent privacy requirements—two critical challenges in real-world healthcare applications. While existing studies on PD classification using semi-supervised learning (SSL) and federated learning (FL) have contributed valuable insights, they remain constrained by several limitations that impact predictive accuracy, generalizability, and practical deployment.

Traditional SSL approaches, especially those relying on handcrafted features and conventional training paradigms, often struggle under label-scarce conditions. For instance, Zhang et al. [21] proposed an

SMC-based framework for PD detection using MRI data, achieving an accuracy of 74%; however, its performance is hindered by limited labeled samples and poor generalization across heterogeneous datasets. Similarly, Sakar et al. [9] reported a maximum accuracy of 89% and an MCC of 59% using feature subset selection, indicating moderate discriminative capability but also revealing challenges related to class imbalance and redundant feature representations. More recent FL-based approaches, such as the federated DL framework introduced by Danek et al. [27], offer advantages in scalability and multi-institutional collaboration. However, these methods often lack explicit privacy guarantees and typically rely on fully labeled datasets or centralized pseudo-labeling strategies, which restrict their effectiveness in low-label settings. Furthermore, baseline SSL methods such as Rasco [60] and RelRasco [61] achieve only moderate accuracy on both Dataset 1 and Dataset 2, underscoring the ongoing difficulty of achieving a balance between robustness, efficiency, and reliable pseudo-label generation.

A key challenge in both SSL and FL lies in achieving balanced performance on highly imbalanced and heterogeneous datasets while preserving strong generalization to unseen data. Many existing methods tend to overfit small labeled subsets or suffer from error propagation due to unreliable pseudo-labels. Additionally, FL frameworks often encounter trade-offs among privacy preservation, model performance, and communication efficiency. In this study, client heterogeneity is realistically modeled using a Dirichlet distribution with a concentration parameter of $\beta = 0.8$, which induces varying class distributions across clients and reflects non-IID clinical data settings. The proposed framework effectively addresses these challenges through iterative refinement of high-confidence pseudo-labels and by leveraging consensus among diverse ensemble learners. This strategy enhances class discrimination without requiring modifications to the underlying model architecture. The effectiveness of the approach is demonstrated through consistent performance gains across both datasets. On Dataset 1, the global model achieves 97.14% accuracy and 97.14% maF1 score, significantly outperforming existing SSL baselines and conventional methods. On Dataset 2, the model attains 95.50% accuracy and 95.50% maF1, indicating improved predictive balance and agreement compared to both SSL and FL benchmarks. Furthermore, validation and test results confirm strong generalization capability, with final model performance exceeding 96% on Dataset 1 and remaining above 93% on Dataset 2. These results highlight the robustness of the proposed framework on unseen data and its effectiveness in mitigating overfitting.

Despite these advancements, several limitations of the proposed framework merit further investigation. The iterative pseudo-labeling strategy, while effective in enhancing label quality and overall performance, introduces additional computational overhead and increases training time compared to lightweight SSL approaches. Although inference remains efficient, this added complexity may pose challenges in resource-constrained environments or real-time deployment scenarios. Moreover, while the framework shows robustness to noise and redundant features, variations across datasets indicate that feature importance does not always generalize consistently. This suggests the need for more adaptive representation learning mechanisms that can dynamically adjust to dataset-specific characteristics and improve generalizability across diverse clinical settings.

There remains substantial scope for further improvement of the proposed framework. Future work can focus on extending the approach to more diverse and complex data settings, as well as exploring more efficient training and optimization strategies to improve computational efficiency and scalability. Additionally, investigating adaptive learning mechanisms may further enhance the model’s ability to generalize across heterogeneous data distributions and effectively handle variations in feature relevance.

5 Conclusion

This work addresses the critical challenge of accurate and privacy-preserving PD classification in scenarios with limited labeled data and decentralized data distributions by integrating FL with an ensemble-guided pseudo-labeling framework. Specifically, XGB, GB, and LGBM models collaboratively generate high-confidence pseudo-labels using a confidence-aware majority voting mechanism. This design creates a dual optimization paradigm that improves label reliability at the local level while enabling robust global knowledge aggregation across heterogeneous clients. Extensive experiments on two benchmark PD speech datasets demonstrate that the proposed framework consistently outperforms conventional centralized and

baseline semi-supervised methods. It achieves superior classification accuracy and maF1 scores under realistic low-label conditions. Notably, the federated global model attains 97.14% accuracy on Dataset 1 and 95.50% on Dataset 2, while exhibiting stable convergence across communication rounds. This indicates effective mitigation of client drift and statistical heterogeneity. Moreover, the federated training paradigm inherently preserves privacy by ensuring that sensitive patient data remains localized. Further analysis using SHAP highlights the contribution of individual speech features, enhancing interpretability and confirming that the framework effectively reduces noise propagation in pseudo-labeled data. As a result, class separability and overall model generalization are significantly improved.

The framework also demonstrates strong practical relevance for real-world healthcare applications. It enables reliable PD detection under limited labeled data while maintaining strict privacy preservation, making it well-suited for integration into clinical workflows and telemedicine systems. This facilitates early diagnosis and continuous patient monitoring, ultimately supporting timely medical intervention and improved patient outcomes. By preserving data locality and safeguarding sensitive information, FEP-SSL addresses key privacy concerns in distributed healthcare environments. Overall, this work advances the state of the art in federated SSL by showing that ensemble-driven pseudo-labeling can effectively bridge the gap between scarce annotations and high predictive performance, providing a scalable, robust, and privacy-aware solution for next-generation intelligent healthcare systems.

Acknowledgments

The authors sincerely acknowledge the Institute of Information Technology (IIT), Jahangirnagar University, Savar, Dhaka, Bangladesh, and the ICT Division fellowship of the Government of The People's Republic of Bangladesh in 2025-2026 (Fiscal year).

Author Contributions

Conceptualization: Md. Shakil Ahmed, Shakila Shafiq, M. Shamim Kaiser.

Data curation: Md. Shakil Ahmed.

Formal analysis: Md. Shakil Ahmed, A. S. M. Sanwar Hosen.

Funding acquisition: M. Shamim Kaiser, A. S. M. Sanwar Hosen.

Investigation: M. Shamim Kaiser, Md. Sazzadur Rahman, Youngbae Hwang.

Methodology: Md. Shakil Ahmed, Shakila Shafiq, M. Shamim Kaiser, Md. Sazzadur Rahman, A. S. M. Sanwar Hosen, Youngbae Hwang.

Resources: Shakila Shafiq, Md. Sazzadur Rahman.

Supervision: M. Shamim Kaiser.

Validation: Md. Shakil Ahmed, Shakila Shafiq, Md. Sazzadur Rahman.

Visualization: Md. Shakil Ahmed, Md. Sazzadur Rahman, A. S. M. Sanwar Hosen, Youngbae Hwang.

Writing – original draft: Md. Shakil Ahmed.

Writing – review & editing: Shakila Shafiq, M. Shamim Kaiser, A. S. M. Sanwar Hosen. y

References

1. Parkinson's Foundation. What is Parkinson's?; 2022. Accessed: Jan. 27, 2022. Available from: <https://www.parkinson.org/understandingparkinsons/what-is-parkinsons>.
2. Noor MBT, Zenia NZ, Kaiser MS, Mahmud M, Mamun SA. Detecting neurodegenerative disease from MRI: A brief review on a deep learning perspective. In: Proceedings of the International Conference on Brain Informatics. Cham, Switzerland: Springer; 2019. p. 115-25.
3. Noor MBT, Zenia NZ, Kaiser MS, Mamun SA, Mahmud M. Application of deep learning in detecting neurological disorders from magnetic resonance images. Brain Informatics. 2020;7(1):1-21.

4. Lacy S. Modeling Movement Disorders in Parkinson's Disease Using Computational Intelligence [PhD thesis]. University of York; 2015.
5. Yurt M, Dalmaz O, Dar S, Ozbey M, Tmaz B, Oguz K, et al. Semi-supervised learning of MRI synthesis without fully-sampled ground truths. *IEEE Transactions on Medical Imaging*. 2022;41(12):3895-906.
6. Yousefirizi F, Shiri I, O JH, Bloise I, Martineau P, Wilson D, et al. Semi-supervised learning towards automated segmentation of PET images with limited annotations: Application to lymphoma patients. *Physical and Engineering Sciences in Medicine*. 2024;47(3):833-49.
7. Eskofier BM, Lee SI, Daneault JF, Golabchi FN, Carvalho GF, Vergara-Diaz G, et al. Recent machine learning advancements in sensor-based mobility analysis. In: *IEEE EMBC*; 2016. p. 655-8.
8. Shafiq S, Ahmed S, Kaiser MS, Mahmud M, Hossain MS, Andersson K. Comprehensive analysis of nature-inspired algorithms for Parkinson's disease diagnosis. *IEEE Access*. 2022;11:1629-53.
9. Sakar CO, Serbes G, Gunduz A, Tunc HC, Nizam H, Sakar BE, et al. A comparative analysis of speech signal processing algorithms for Parkinson's disease classification. *Applied Soft Computing*. 2019;74:255-63.
10. Al Banna M, Ghosh T, Taher KA, Kaiser MS, Mahmud M. A monitoring system for patients of autism spectrum disorder using artificial intelligence. In: *International Conference on Brain Informatics*; 2020. p. 251-62.
11. Jesmin S, Kaiser MS, Mahmud M. Artificial and Internet of Healthcare things based Alzheimer care during COVID 19. In: *International Conference on Brain Informatics*; 2020. p. 263-74.
12. Ahmed S, Hossain M, Nur SB, Kaiser MS, Mahmud M. Toward machine learning-based psychological assessment of autism spectrum disorders. In: *Trends in Electronic Health Informatics*; 2022. p. 139-49.
13. Mahmud M, Kaiser MS, Rahman MA, et al. Towards explainable and privacy-preserving AI for autism spectrum disorder. In: *Human-Computer Interaction Conference*; 2022. p. 356-70.
14. Nahiduzzaman M, Tasnim M, Newaz NT, Kaiser MS, Mahmud M. Machine learning based early fall detection for elderly people. In: *International Conference on Brain Informatics*; 2020. p. 204-14.
15. Biswas M, Rahman A, Kaiser MS, Mamun SA, et al. Indoor navigation support system for patients with neurodegenerative diseases. In: *International Conference on Brain Informatics*; 2021. p. 411-22.
16. Mahmud M, Kaiser MS, Hussain A, Vassanelli S. Applications of deep learning and reinforcement learning to biological data. *IEEE Transactions on Neural Networks and Learning Systems*. 2018;29(6):2063-79.
17. Ahmed S, Hossain M, Amit AA, Tahsin M, Mahmud M, Kaiser MS. Real-time Bangla sign language detection and recognition using YOLOv10. In: *Trends in Computational and Cognitive Engineering*. Springer; 2023. p. 383-400.
18. Hady MFA, Schwenker F. Semi-supervised learning. In: *Handbook on Neural Information Processing*. Springer; 2013. p. 215-39.
19. Zhao Y, Li M, Lai L, Suda N, Civin D, Chandra V. Federated learning with non-IID data. *arXiv preprint arXiv:1806.00582*. 2018.
20. Li X, Huang K, Yang W, Wang S, Zhang Z. On the convergence of FedAvg on non-IID data. *arXiv preprint arXiv:1907.02189*. 2019.
21. Zhang X, Zhai D, Yang Y, Zhang Y, Wang C. A Novel Semi-Supervised Multi-View Clustering Framework for Parkinson's Disease Screening. *arXiv preprint arXiv:2003.04760*. 2020.

22. Ammour A, Aouraghe I, Khaissidi G, Mrabti M, Aboulem G, Belahsen F. A Semi-Supervised Approach for Characterizing Arabic Online Handwriting in Parkinson's Disease Patients. *Computer Methods and Programs in Biomedicine*. 2020;183:104979. doi:10.1016/j.cmpb.2019.104979.
23. Bao G, Lin M, Sang X, Hou Y, Liu Y, Wu Y. Classification of Dysphonic Voice Signals in Parkinson's Disease Using a Semi-Supervised Competitive Learning Algorithm. *Biosensors*. 2022;12(7):502.
24. Malekroodi HS, Madusanka N, Lee Bi, Yi M. Speech-Based Parkinson's Detection Using Pre-Trained Self-Supervised Automatic Speech Recognition Models and Supervised Contrastive Learning. *Bioengineering*. 2025;12(7):728. doi:10.3390/bioengineering12070728.
25. La Quatra M, Orozco-Arroyave JR, Siniscalchi MS. Bilingual Dual-Head Deep Model for Parkinson's Disease Detection from Speech. *arXiv preprint arXiv:250310301*. 2025.
26. Sarlas A, Kalafatelis A, Alexandridis G, Kourtis MA, Trakadas P. Exploring Federated Learning for Speech-Based Parkinson's Disease Detection. In: *Proceedings of the 18th International Conference on Availability, Reliability and Security (ARES)*; 2023. p. 1-6.
27. Danek BP, Makarious MB, Dadu A, Vitale D, Lee PS, Singleton AB, et al. Federated Learning for Multi-Omics: A Performance Evaluation in Parkinson's Disease. *Patterns*. 2024;5(3).
28. Pang M, Xu H, Huang Z, Zhou Y, Chen S, Wang B, et al. A Tri-Factor Adaptive Federated Learning Framework for Parkinson's Disease Diagnosis via Multi-Source Facial Expression Analysis. *IEEE Journal of Biomedical and Health Informatics*. 2025.
29. Dipro SH, Islam M, Al Nahian A, Azad MS, Chakrabarty A, Reza T. A Federated Learning-Based Privacy-Preserving Approach for Detecting Parkinson's Disease Using Deep Learning. In: *Proceedings of the 25th International Conference on Computer and Information Technology (ICCIT)*; 2022. p. 139-44.
30. Kaissis GA, Makowski MR, Rückert D, Braren RF. Secure, Privacy-Preserving and Federated Machine Learning in Medical Imaging. *Nature Machine Intelligence*. 2020;2(6):305-11.
31. Ogundokun RO, Misra S, Maskeliunas R, Damasevicius R. A Review on Federated Learning and Machine Learning Approaches: Categorization, Application Areas, and Blockchain Technology. *Information*. 2022;13(5):263.
32. Ali M, Naeem F, Tariq M, Kaddoum G. Federated Learning for Privacy Preservation in Smart Healthcare Systems: A Comprehensive Survey. *IEEE Journal of Biomedical and Health Informatics*. 2022;27(2):778-89.
33. Qiu L, Cheng J, Gao H, Xiong W, Ren H. Federated Semi-Supervised Learning for Medical Image Segmentation via Pseudo-Label Denoising. *IEEE Journal of Biomedical and Health Informatics*. 2023;27(10):4672-83.
34. Fares MH, Saad AMS. Towards Privacy-Preserving Medical Imaging: Federated Learning with Differential Privacy and Secure Aggregation Using a Modified ResNet Architecture. *Journal of Medical Imaging*. 2024;12:119-30. doi:10.1007/s12345-025-01025-9.
35. Tanim SA, Aurnob AR, Mridha MF, Safran M, Alfarhood S, Che D. Secure Federated Learning for Parkinson's Disease: Non-IID Data Partitioning and Homomorphic Encryption Strategies. *IEEE Access*. 2024. doi:10.1109/ACCESS.2024.3454690.
36. Kaur S, Sankar R. Integrative Federated Learning Framework for Multimodal Parkinson's Disease Biomarker Fusion. *Medical Measurement and Computing*. 2024. doi:10.1007/s11517-024-03139-3.
37. Sakar CO, Serbes G, Gunduz A, Nizam H, Sakar BE. Parkinson's Disease Classification; 2018. DOI: <https://doi.org/10.24432/C5MS4X>. UCI Machine Learning Repository.

38. Prez C. Parkinson Dataset with Replicated Acoustic Features; 2016. UCI Machine Learning Repository. Available from: <https://doi.org/10.24432/C5701F>.
39. Sakar CO, Serbes G, Gunduz A, Tunc HC, Nizam H, Sakar BE, et al. A Comparative Analysis of Speech Signal Processing Algorithms for Parkinson's Disease Classification and the Use of the Tunable Q-Factor Wavelet Transform. *Applied Soft Computing*. 2019;74:255-63.
40. Murty KSR, Yegnanarayana B. Combining evidence from residual phase and MFCC features for speaker recognition. *IEEE Signal Processing Letters*. 2006;13(1):52-5.
41. Godino-Llorente JJ, Gomez-Vilda P, Blanco-Velasco M. Dimensionality reduction of a pathological voice quality assessment system based on Gaussian mixture models and short-term cepstral parameters. *IEEE Transactions on Biomedical Engineering*. 2006;53(10):1943-53.
42. Peker M. A decision support system to improve medical diagnosis using a combination of K-medoids clustering based attribute weighting and SVM. *Journal of Medical Systems*. 2016;40(5):116.
43. Tufekci Z, Gowdy JN. Feature extraction using discrete wavelet transform for speech recognition. In: *Proceedings of IEEE SoutheastCon 2000, Preparing for the New Millennium*. IEEE; 2000. p. 116-23.
44. Prez C. UCI, editor. Parkinson Dataset with Replicated Acoustic Features. UCI Machine Learning Repository; 2016. Available from: <https://archive.ics.uci.edu/dataset/489/parkinson+dataset+with+replicated+acoustic+features>. doi:10.24432/C5701F.
45. Naranjo L, Perez CJ, Campos-Roca Y, Martin J. Addressing Voice Recording Replications for Parkinson's Disease Detection. *Expert Systems with Applications*. 2016;46:286-92.
46. Naranjo L, Perez CJ, Martin J, Campos-Roca Y. A Two-Stage Variable Selection and Classification Approach for Parkinson's Disease Detection by Using Voice Recording Replications. *Computer Methods and Programs in Biomedicine*. 2017;142:147-56.
47. Jia BB, Zhang ML. Multi-Dimensional Classification via Sparse Label Encoding. In: *Proc. ICML*; 2021. p. 4917-26.
48. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*. 2002;16:321-57.
49. Chen T, He T, Benesty M, Khotilovich V, Tang Y, Cho H, et al. XGBoost: Extreme Gradient Boosting. R package version 04-2. 2015.
50. Ayyadevara VK. Gradient Boosting Machine. In: *Pro Machine Learning Algorithms*. Apress; 2018. p. 117-34.
51. Ke G, Meng Q, Finley T, Wang T, Chen W, Ma W, et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In: *Advances in Neural Information Processing Systems*; 2017. p. 3149-57.
52. Donepudi S, Nakka R, Thota KK, Ajmeera M, Praveen SP, Sindhura S. Enhancing Person-Centric Health Care for Diabetes Prediction. In: *Ambient Assistive Technology*, Vol. 2. Springer; 2025. p. 127-55.
53. Lee DH, et al. Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks. In: *Workshop on challenges in representation learning, ICML*. Atlanta; 2013. p. 896.
54. Zhu X, Goldberg AB. *Introduction to Semi-Supervised Learning*. Morgan & Claypool; 2009.
55. Zhou ZH, Li M. Tri-Training: Exploiting Unlabeled Data Using Three Classifiers. *IEEE Transactions on Knowledge and Data Engineering*. 2005;17(11):1529-41.
56. Li T, Sahu AK, Zaheer M, Sanjabi M, Talwalkar A, Smith V. Federated optimization in heterogeneous networks. *Proceedings of Machine learning and systems*. 2020;2:429-50.

57. Jahin MA, Masud MA, Mridha MF, Aung Z, Dey N. KACQ-DCNN: Uncertainty-Aware Interpretable Kolmogorov–Arnold Classical–Quantum Dual-Channel Neural Network for Heart Disease Detection. *Computers in Biology and Medicine*. 2025;197:110976.
58. Sakar CO, Serbes G, Gunduz A, Tunc HC, Nizam H, Sakar BE, et al. A Comparative Analysis of Speech Signal Processing Algorithms for Parkinson’s Disease Classification and the Use of the Tunable Q-Factor Wavelet Transform. *Applied Soft Computing*. 2019;74:255-63.
59. Garrido-Labrador JL, Maudes-Raedo JM, Rodríguez JJ, García-Osorio CI. SSLearn: A Semi-Supervised Learning Library for Python. *SoftwareX*. 2025;29:102024.
60. Wang J, Luo SW, Zeng XH. A Random Subspace Method for Co-Training. In: *Proceedings of the IEEE International Joint Conference on Neural Networks*; 2008. p. 195-200. doi:10.1109/IJCNN.2008.4633789.
61. Yaslan Y, Cataltepe Z. Co-Training with Relevant Random Subspaces. *Neurocomputing*. 2010;73(10–12):1652-61. doi:10.1016/j.neucom.2010.01.018.
62. Hady MFA, Schwenker F. Co-Training by Committee: A New Semi-Supervised Learning Framework. In: *Proceedings of IEEE International Conference on Data Mining Workshops*; 2008. p. 563-72. doi:10.1109/ICDMW.2008.27.
63. Blum A, Mitchell T. Combining Labeled and Unlabeled Data with Co-Training. In: *Proceedings of the Eleventh Annual Conference on Computational Learning Theory*; 1998. p. 92-100. doi:10.1145/279943.279962.
64. Li M, Zhou ZH. SETRED: Self-Training with Editing. In: Ho TB, Cheung D, Liu H, editors. *Advances in Knowledge Discovery and Data Mining*. Springer; 2005. p. 611-21. doi:10.1007/1143091971.
65. Deng C, Guo MZ. Tri-Training and Data Editing Based Semi-Supervised Clustering Algorithm. In: *MICAI 2006: Advances in Artificial Intelligence*. Springer; 2006. p. 641-51. doi:10.1007/1192523161.
66. Li M, Zhou ZH. Improve Computer-Aided Diagnosis with Machine Learning Techniques Using Undiagnosed Samples. *IEEE Transactions on Systems, Man, and Cybernetics, Part A*. 2007;37(6):1088-98. doi:10.1109/TSMCA.2007.904745.
67. Zhou ZH, Li M. Tri-Training: Exploiting Unlabeled Data Using Three Classifiers. *IEEE Transactions on Knowledge and Data Engineering*. 2005;17(11):1529-41. doi:10.1109/TKDE.2005.186.
68. Zhou Y, Goldman S. Democratic Co-Learning. In: *Proceedings of the 16th IEEE International Conference on Tools with Artificial Intelligence*; 2004. p. 594-602. doi:10.1109/ICTAI.2004.48.
69. Yarowsky D. Unsupervised Word Sense Disambiguation Rivaling Supervised Methods. In: *Proceedings of the 33rd Annual Meeting of the Association for Computational Linguistics*; 1995. p. 189-96. doi:10.3115/981658.981684.
70. Adnan T, Abdelkader A, Liu Z, Hossain E, Park S, Islam MD, et al. A Novel Fusion Architecture for PD Detection Using Semi-Supervised Speech Embeddings. *arXiv preprint arXiv:240517206*. 2024.
71. Egbo B, et al. Explainable Machine Learning for Early Detection of Parkinson’s Disease. *Frontiers in Aging Neuroscience*. 2025;17:1672971. doi:10.3389/fnagi.2025.1672971.